



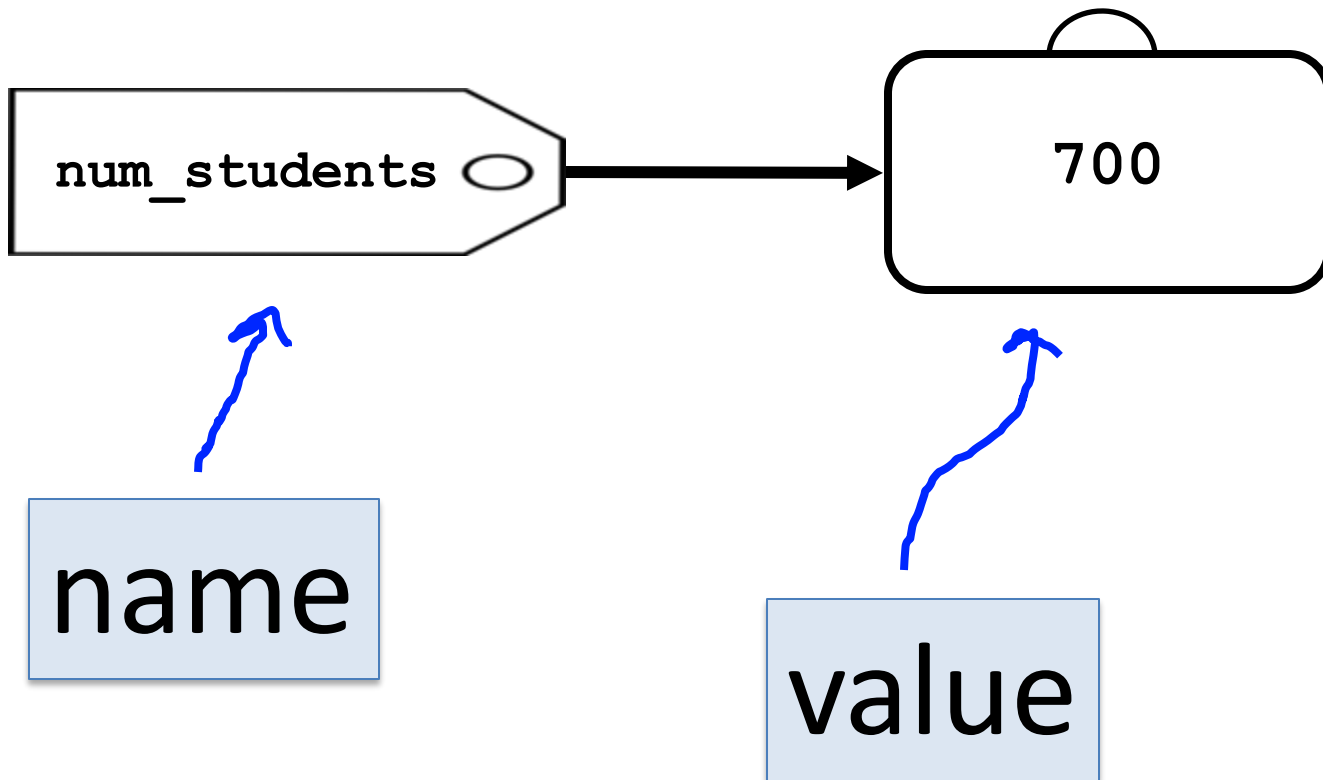
While/If Revisited

Chris Piech and Mehran Sahami
Stanford University

Review

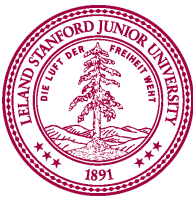
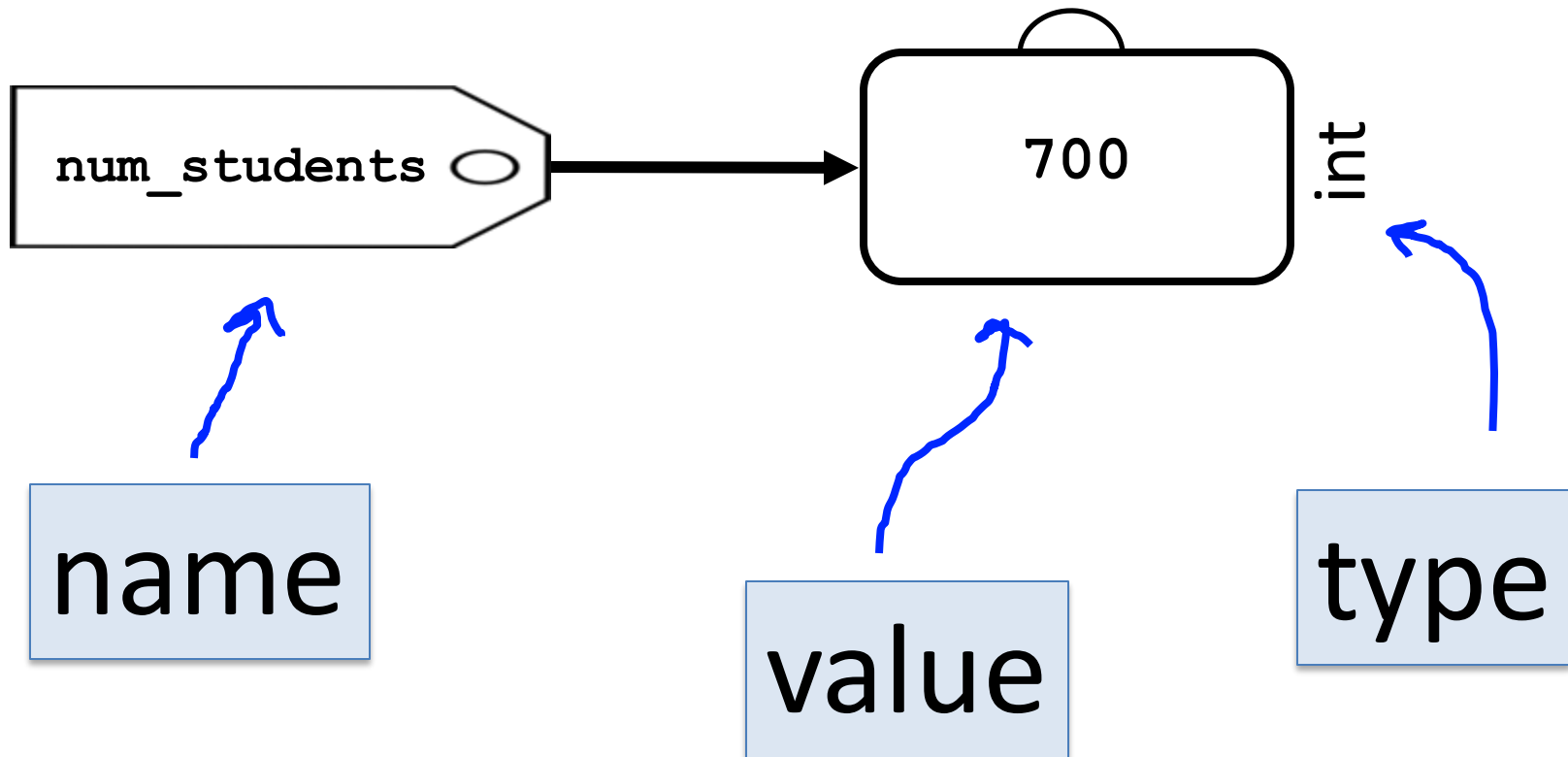
Variables are like Boxes

```
num_students = 700
```

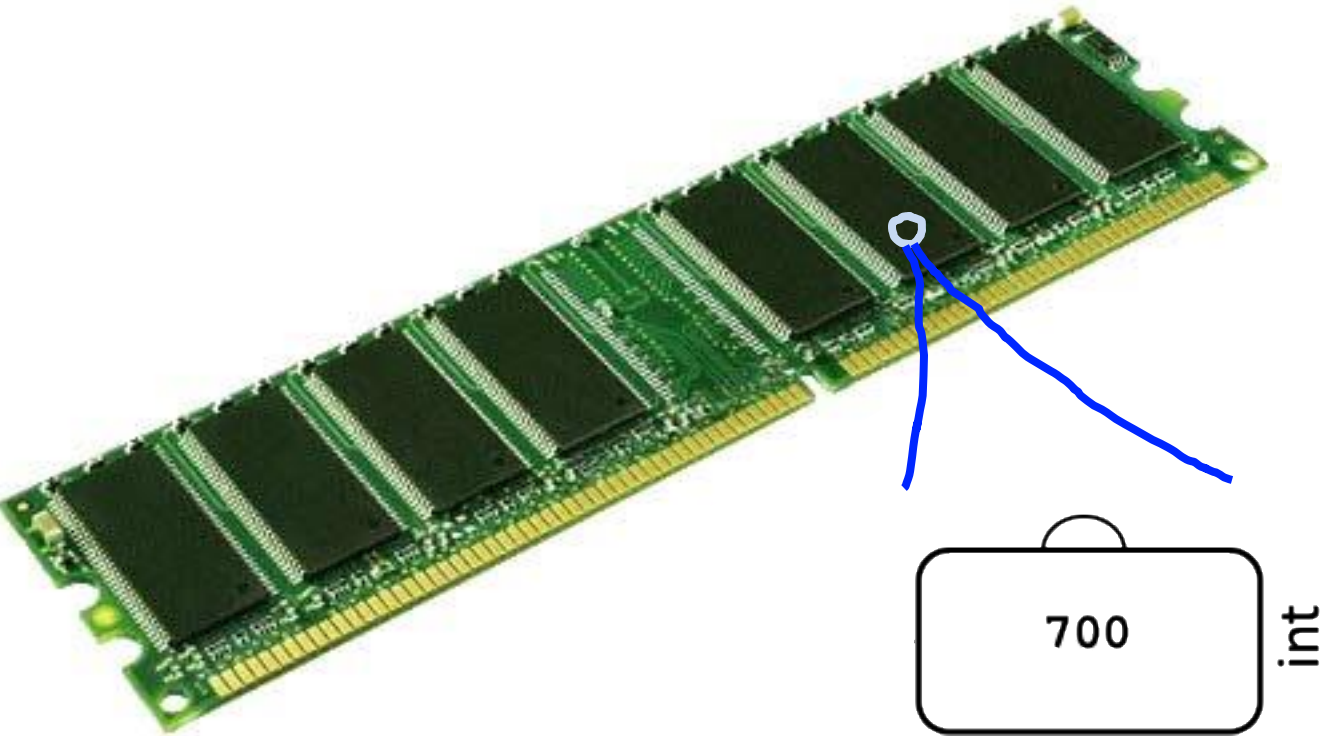


Variables are like Boxes

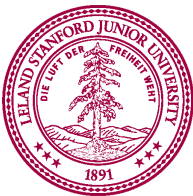
```
num_students = 700
```



Teeny Tiny Boxes



My computer has space for
about 10 billion boxes



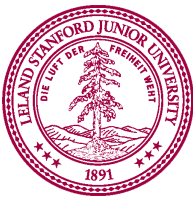
Create, Modify, Use

```
# Create a variable, of type int  
# called age.
```

```
age = 37
```

```
# Use the value in age (output it)  
print(f"age is: {age}")
```

```
# Modify age to be one greater.  
age = age + 1
```



Create, Modify, Use

Create a variable, of type int

called age.

```
age = 37
```

Use the value in age (output it)

```
print(f"age is: {age}")
```

Modify age to be one greater.

```
age = age + 1
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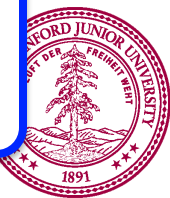
Arithmetic Operators

+ Addition

* Multiplication

- Subtraction

/ Division

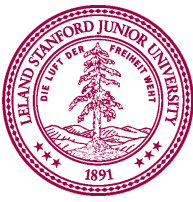


Review Example: Bolt Calculator



Usain Bolt has the record speed for any human

He was recorded going 10.44 meters per second



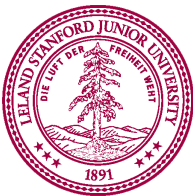
Review: Bolt Calculator

SPEED = 10.44 # Bolt's speed in meters / second

```
def main():  
    time_str = input("Run time (s): ")  
    time = float(time_str)  
    distance = SPEED * time  
    print(f"Bolt can run {distance} meters.")
```

terminal

```
% python main.py
```



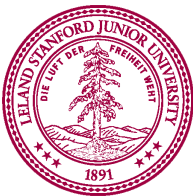
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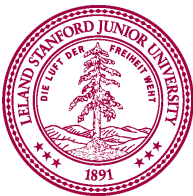
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    distance = SPEED * time  
    print(f"Bolt can run {distance} meters.")
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terminal

```
% python main.py
```



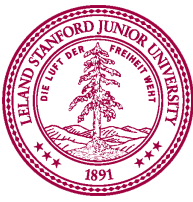
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    time = float(time_str)  
    distance = SPEED * time  
    print(f"Bolt can run {distance} meters.")
```

terminal

```
% python main.py  
Run time (s):
```



Review: Bolt Calculator

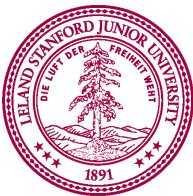
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    time = float(time_str)  
    distance = SPEED * time  
    print(f"Bolt can run {distance} meters.")
```

terminal

```
% python main.py
```

```
Run time (s):
```



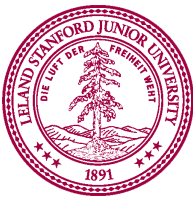
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def main():  
    time_str = input("Run time (s): ")  
    time = float(time_str)  
    distance = SPEED * time  
    print(f"Bolt can run {distance} meters.")
```

terminal

```
% python main.py  
Run time (s): 60
```



Review: Bolt Calculator

SPEED = 10.44 # Bolt's speed in meters / second

def main():

time_str = input("Run time (s): ")

time = float(time_str)

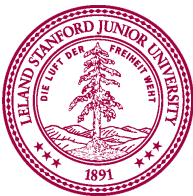
distance = SPEED * time

print(f"Bolt can run {distance} meters.")

terminal

```
% python main.py
```

```
Run time (s): 60
```



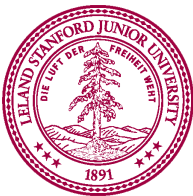
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```
def main():  
    time_str = input("Run time (s): ")  
    time = float(time_str)  
    distance = SPEED * time  
    print(f"Bolt can run {distance} meters.")
```

terminal

```
% python main.py  
Run time (s): 60
```



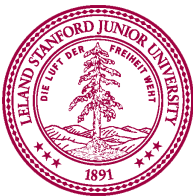
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```
def main():  
    time_str = input("Run time (s): ")  
    time = float(time_str)  
    distance = SPEED * time  
    print(f"Bolt can run {distance} meters.")
```

terminal

```
% python main.py  
Run time (s): 60
```



Review: Bolt Calculator

SPEED = 10.44 # Bolt's speed in meters / second

```
def main():
```

```
    time_str = input("Run time (s): ")
```

```
    time = float(time_str)
```

```
    distance = SPEED * time
```

```
    print(f"Bolt can run {distance} meters.")
```

626.4

terminal

```
% python main.py
```

```
Run time (s): 60
```

str

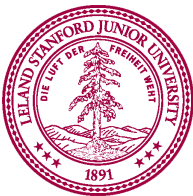
"60"

time_str

float

60.0

time



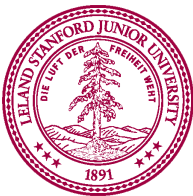
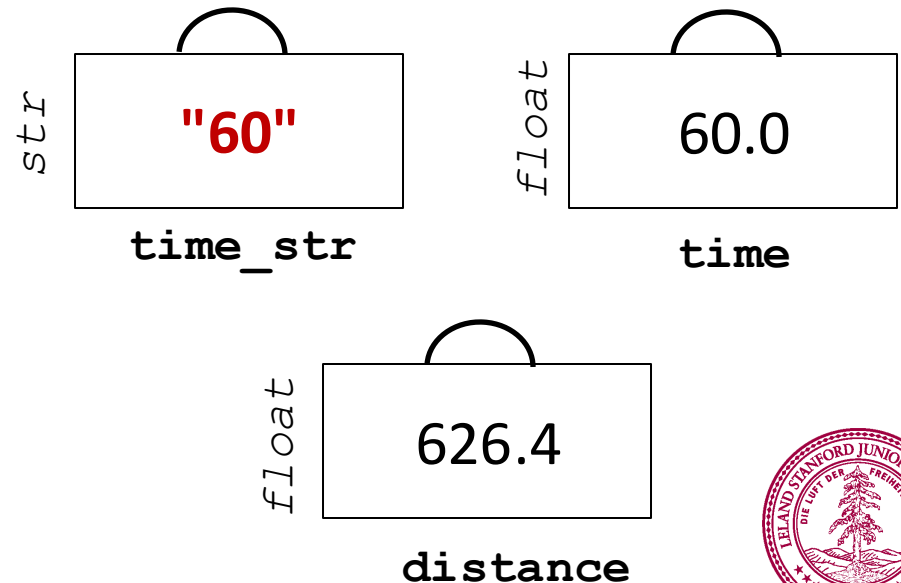
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```
def main():  
    time_str = input("Run time (s): ")  
    time = float(time_str)  
    distance = SPEED * time  
    print(f"Bolt can run {distance} meters.")
```

terminal

```
% python main.py  
Run time (s): 60
```



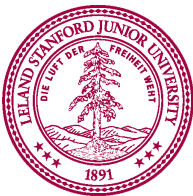
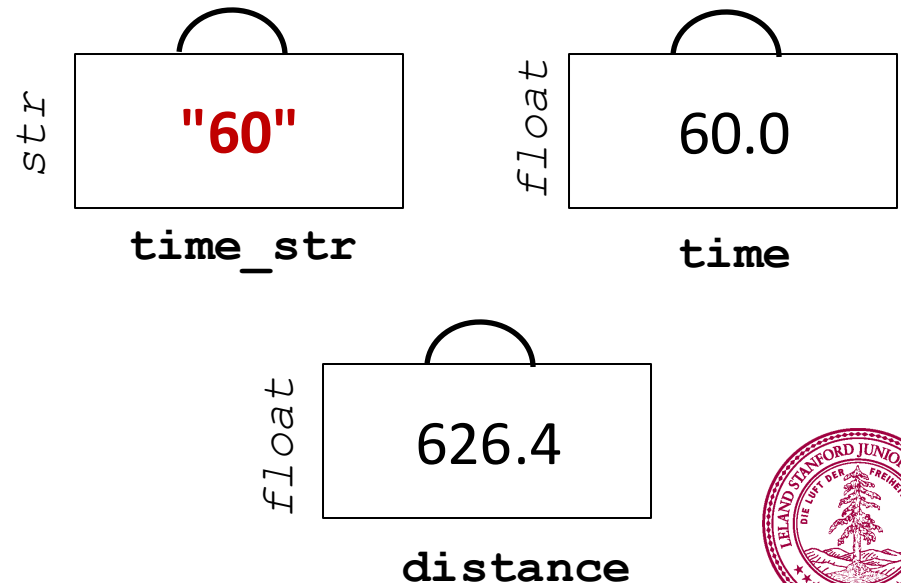
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    time = float(time_str)  
    distance = SPEED * time  
    print(f"Bolt can run {distance} meters.")
```

terminal

```
% python main.py  
Run time (s): 60
```



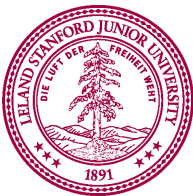
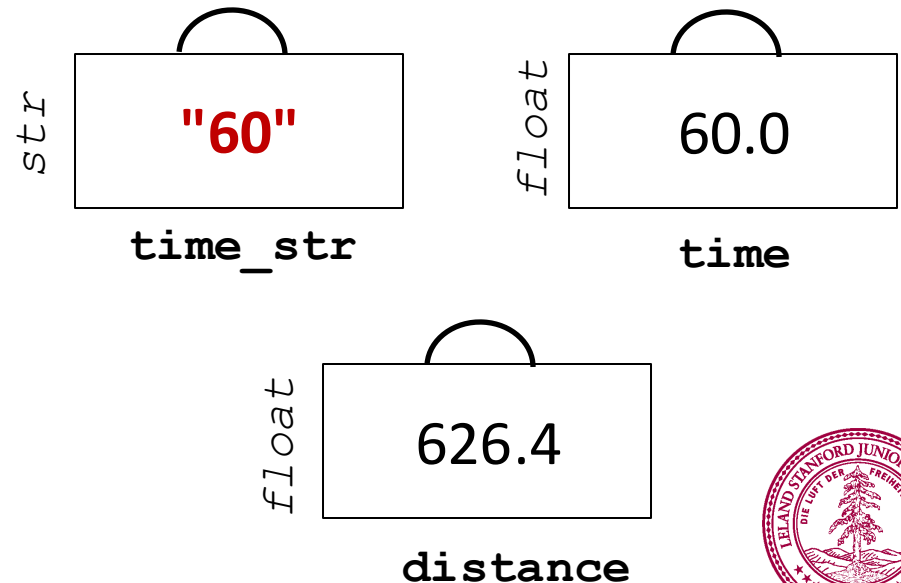
Review: Bolt Calculator

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```
def main():  
    time_str = input("Run time (s): ")  
    time = float(time_str)  
    distance = SPEED * time  
    print(f"Bolt can run {distance} meters.")
```

terminal

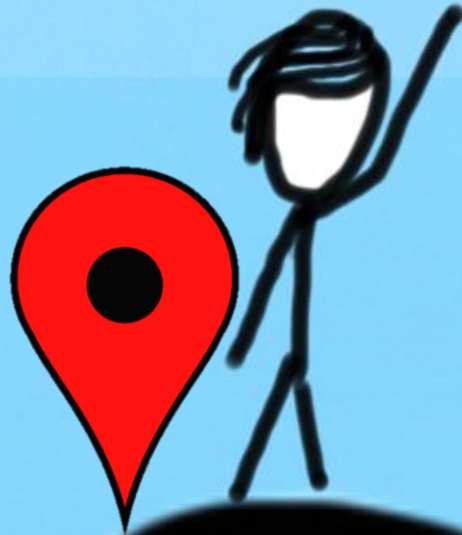
```
% python main.py  
Run time (s): 60  
Bolt can run 626.4 meters
```



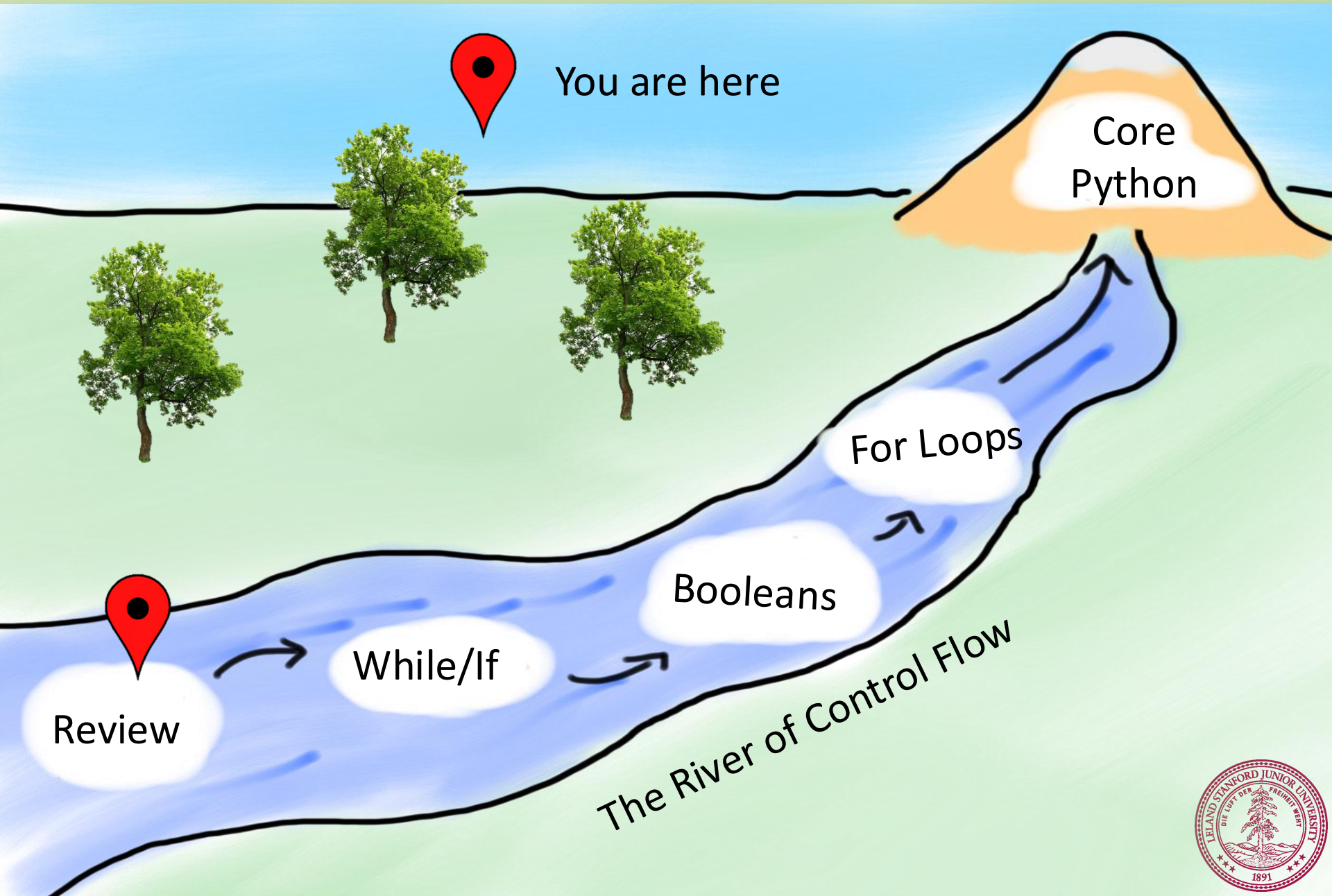
End Review

Today's Goal

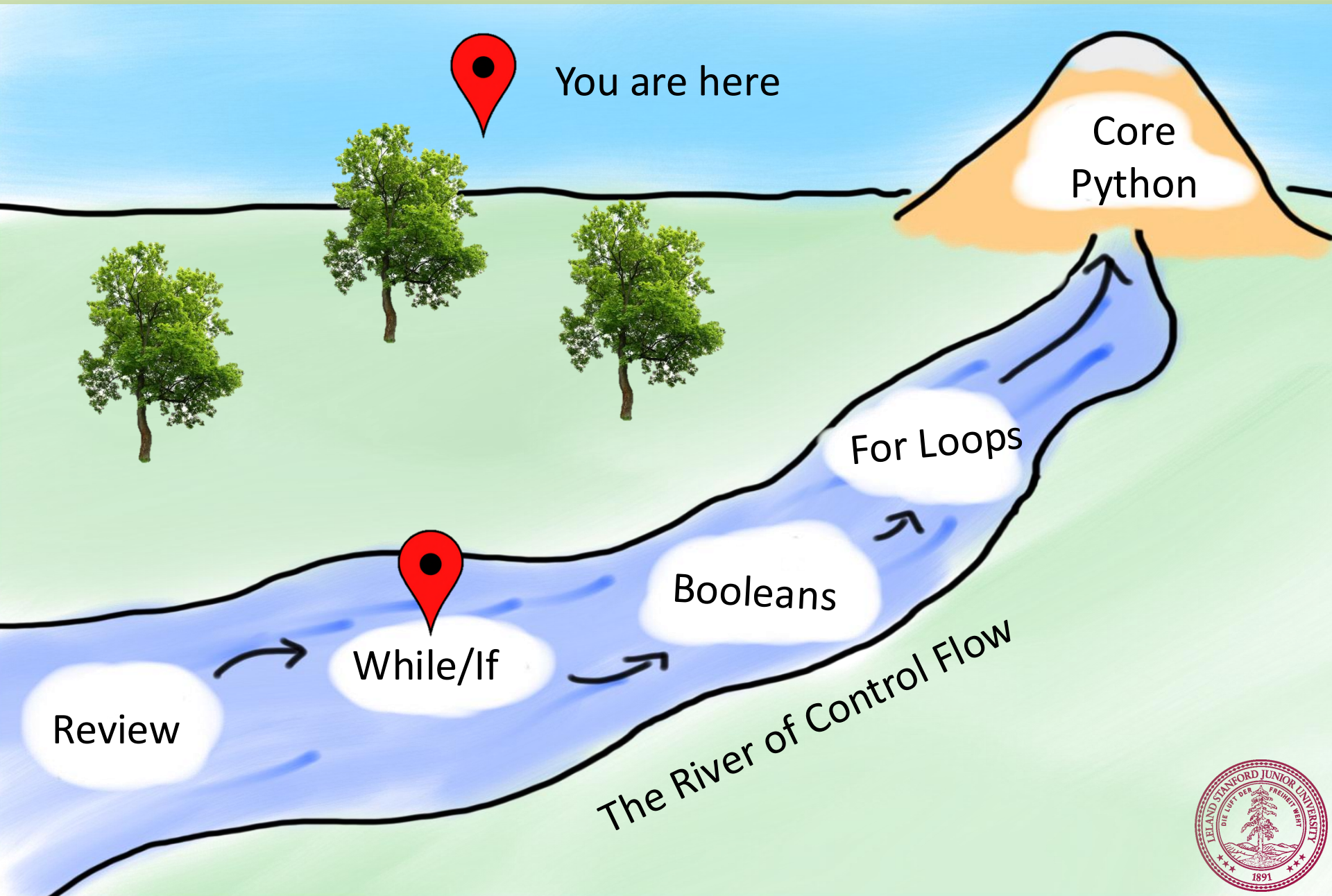
1. Be able to use For / While / If in Python



Today's Route



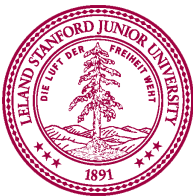
Today's Route



While Loop in Karel

```
while front_is_clear() :  
    body
```

```
if beepers_present() :  
    body
```

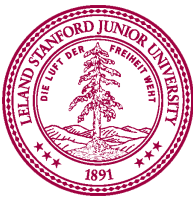


While Loop Redux

while *condition* :
 body

if *condition* :
 body

The condition should be a **boolean** which is
either **True** or **False**



While Loop: Bolt Distance

```
SPEED = 10.44    # Bolt's speed in meters / second
```

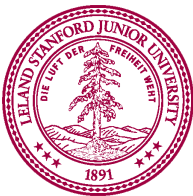
```
def main():
```

```
    time_str = input("Run time (s): ")
```

```
    time = float(time_str)
```

```
    distance = SPEED * time
```

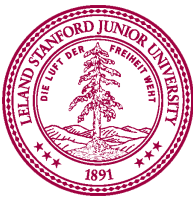
```
    print(f"Bolt can run {distance} meters.")
```



While Loop: Bolt Distance

```
SPEED = 10.44    # Bolt's speed in meters / second
```

```
def main():  
    while True:  
        time_str = input("Run time (s): ")  
        time = float(time_str)  
        distance = SPEED * time  
        print(f"Bolt can run {distance} meters.")
```



While Loop: Bolt Distance

```
SPEED = 10.44    # Bolt's speed in meters / second
```

```
def main():
```

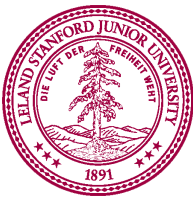
```
    while True:
```

```
        time_str = input("Run time (s): ")
```

```
        time = float(time_str)
```

```
        distance = SPEED * time
```

```
        print(f"Bolt can run {distance} meters.")
```



While Loop: Bolt Distance

```
SPEED = 10.44    # Bolt's speed in meters / second
```

```
def main():
```

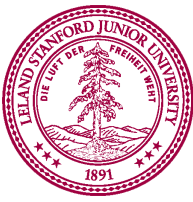
```
    while True:
```

```
        time_str = input("Run time (s): ")
```

```
        time = float(time_str)
```

```
        distance = SPEED * time
```

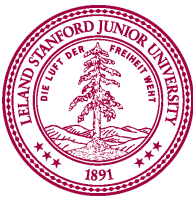
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        print(f"Bolt can run {distance} meters.")
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While Loop: Bolt Distance

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        distance = SPEED * time  
        print(f"Bolt can run {distance} meters.")
```



While Loop: Bolt Distance

```
SPEED = 10.44    # Bolt's speed in meters / second
```

```
def main():
```

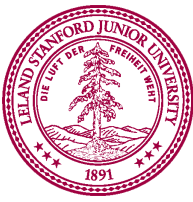
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```

```
        time_str = input("Run time (s): ")
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```
        time = float(time_str)
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```
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```
        print(f"Bolt can run {distance} meters.")
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While Loop: Bolt Distance

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```
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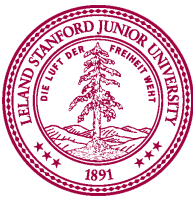
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```
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```

```
        print(f"Bolt can run {distance} meters.")
```



While Loop: Bolt Distance

```
SPEED = 10.44    # Bolt's speed in meters / second
```

```
def main():
```

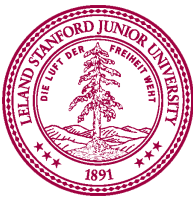
```
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```

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```
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```
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```

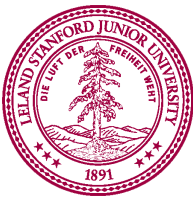
```
        print(f"Bolt can run {distance} meters.")
```



While Loop: Bolt Distance

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def main():  
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While Loop: Bolt Distance

```
SPEED = 10.44    # Bolt's speed in meters / second
```

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def main():
```

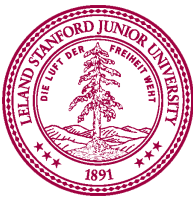
```
    while True:
```

```
        time_str = input("Run time (s): ")
```

```
        time = float(time_str)
```

```
        distance = SPEED * time
```

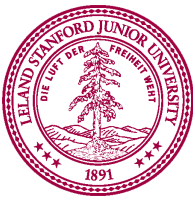
```
        print(f"Bolt can run {distance} meters.")
```



While Loop: Bolt Distance

SPEED = 10.44 # Bolt's speed in meters / second

```
def main():  
    while True:   
        time_str = input("Run time (s): ")  
        time = float(time_str)  
        distance = SPEED * time  
        print(f"Bolt can run {distance} meters.")
```



While Loop: Bolt Distance

```
SPEED = 10.44    # Bolt's speed in meters / second
```

```
def main():
```

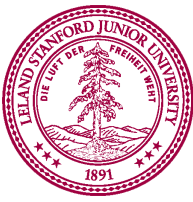
```
    while True:
```

```
        time_str = input("Run time (s): ")
```

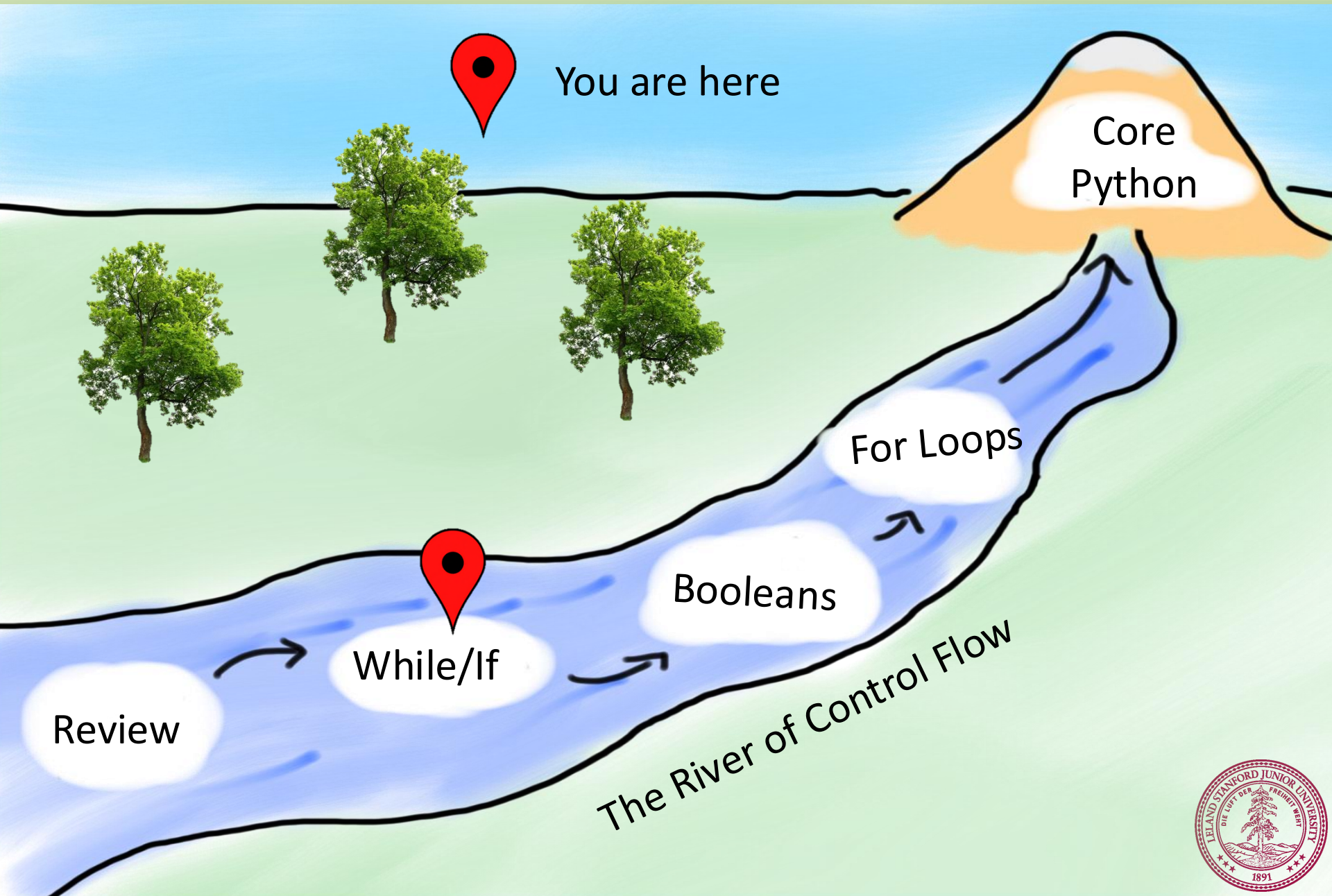
```
        time = float(time_str)
```

```
        distance = SPEED * time
```

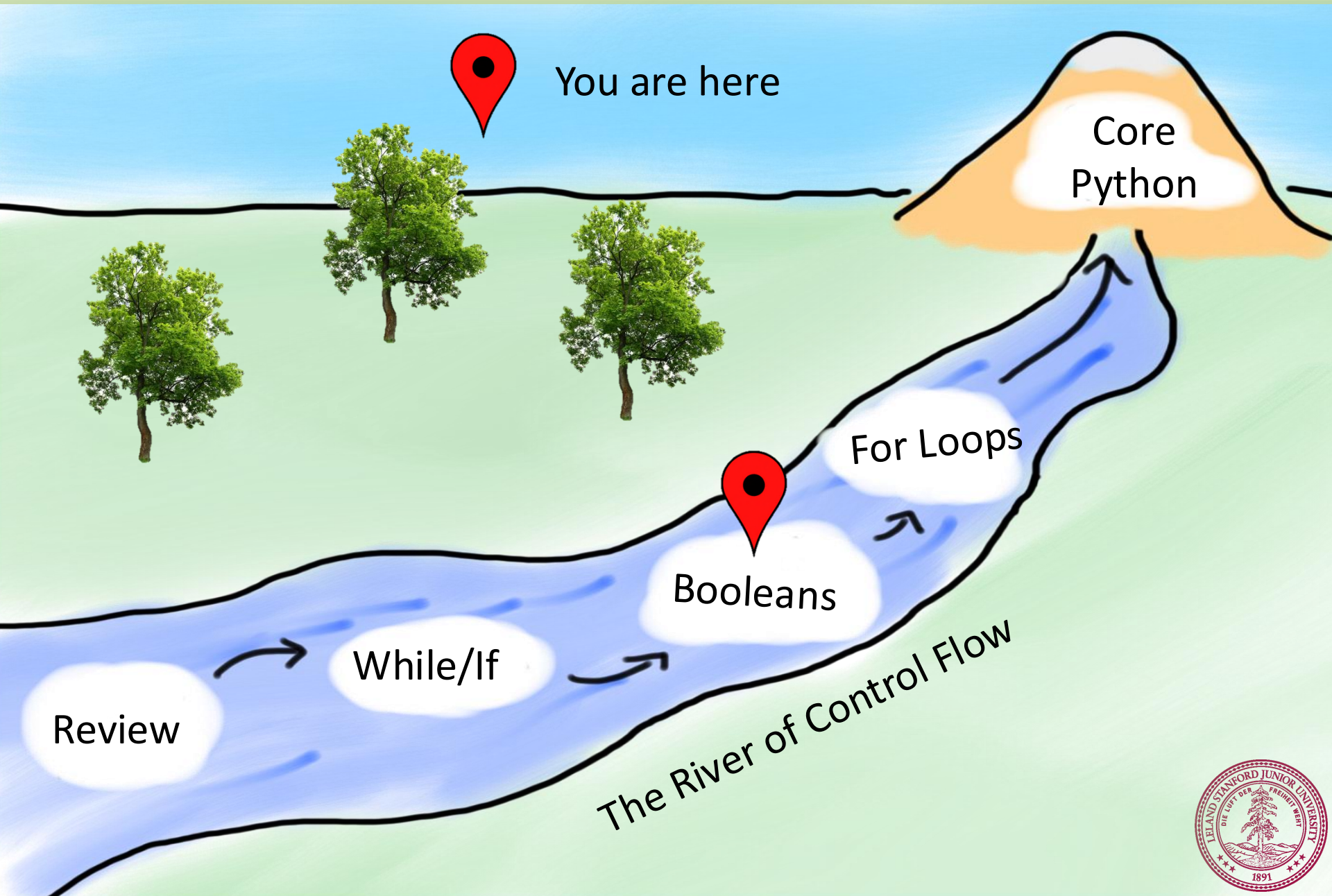
```
        print(f"Bolt can run {distance} meters.")
```



Today's Route



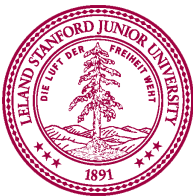
Today's Route



Booleans

`front_is_clear()`

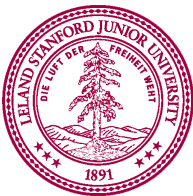
True



Booleans

beepers_present()

True

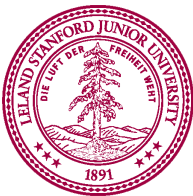


Booleans

`s = "123"`

`s.isdigit()`

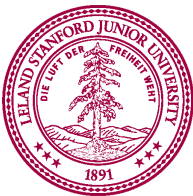
True



Booleans

`my_var < 3`

True



Comparison Operators

Operator	Meaning	Example	Value
==	equals	$1 + 1 == 2$	True
!=	does not equal	$3.2 != 2.5$	True
<	less than	$10 < 5$	False
>	greater than	$10 > 5$	True
<=	less than or equal to	$126 <= 100$	False
>=	greater than or equal to	$5.0 >= 5.0$	True

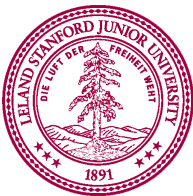
* All have equal precedence



Comparison Operators

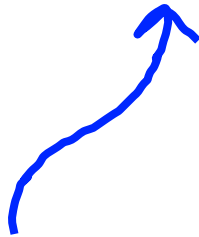
Operator	Meaning	Example	Value
==	equals	1 + 1 == 2	True
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<	less than	10 < 5	False
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>=	greater than or equal to	5.0 >= 5.0	True

* All have equal precedence



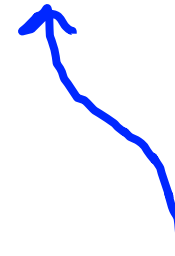
Spot the difference #1

$x = 7$

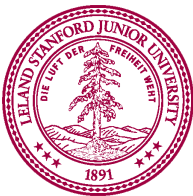


Sets the value of a variable named x to be 7. Creates the variable if it didn't exist.

$x == 7$

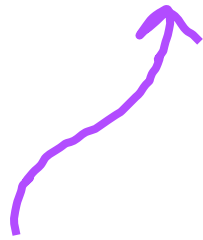


Checks if a variable named x has the value 7



Spot the difference #2

`x == 5`

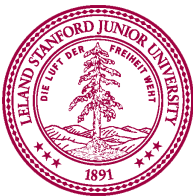


Checks if x is the
number 5

`x == "5"`



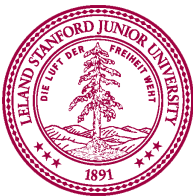
Checks if x is the
string 5



Comparison Operators

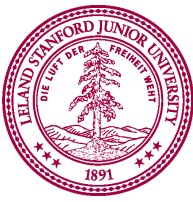
```
if 1 < 2 :  
    print("1 is less than 2")
```

```
num = int(input("Enter a number: "))  
if num == 0:  
    print("That number is 0")  
else :  
    print("That number is not 0.")
```



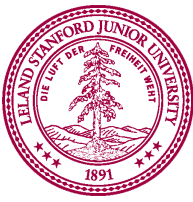
If Else Revisited

```
num = int(input("Enter a number: "))  
if num == 0:  
    print("Your number is 0 ")  
else:  
    if num > 0:  
        print("Your number is positive")  
    else:  
        print("Your number is negative")
```



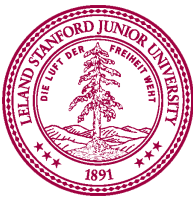
If Else Revisited

```
num = int(input("Enter a number: "))  
if num == 0:  
    print("Your number is 0 ")  
elif num > 0:  
    print("Your number is positive")  
else:  
    print("Your number is negative")
```



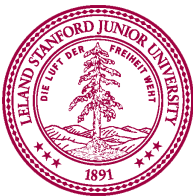
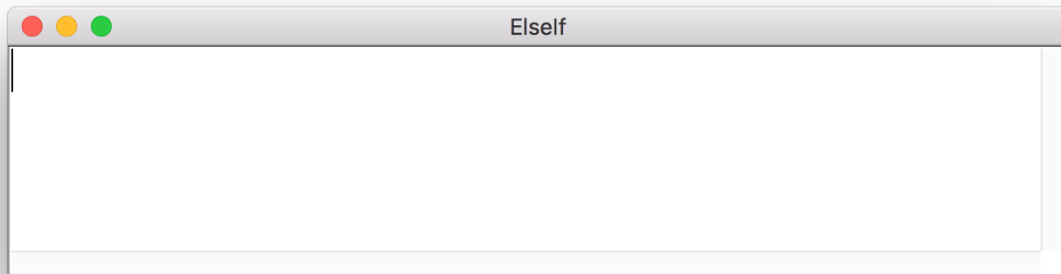
If Else Revisited

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num = int(input("Enter a number: "))  
if num == 0:  
    print("Your number is 0 ")  
elif num > 0:  
    print("Your number is positive")  
else:  
    print("Your number is negative")
```



If Else Revisited

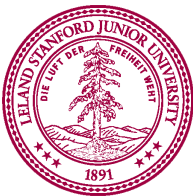
```
num = int(input("Enter a number: "))  
if num == 0:  
    print("Your number is 0 ")  
elif num > 0:  
    print("Your number is positive")  
else:  
    print("Your number is negative")
```



If Else Revisited

"5"

```
num = int(input("Enter a number: "))  
if num == 0:  
    print("Your number is 0 ")  
elif num > 0:  
    print("Your number is positive")  
else:  
    print("Your number is negative")
```



If Else Revisited

5

"5"

```
num = int(input("Enter a number: "))
```

```
if num == 0:
```

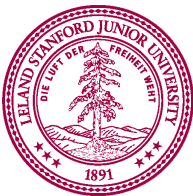
```
    print("Your number is 0 ")
```

```
elif num > 0:
```

```
    print("Your number is positive")
```

```
else:
```

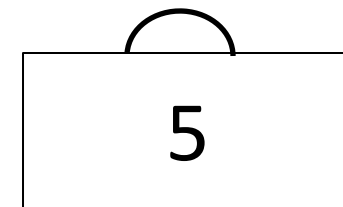
```
    print("Your number is negative")
```



If Else Revisited

5

```
num = int(input("Enter a number: "))  
if num == 0:  
    print("Your number is 0 ")  
elif num > 0:  
    print("Your number is positive")  
else:  
    print("Your number is negative")
```

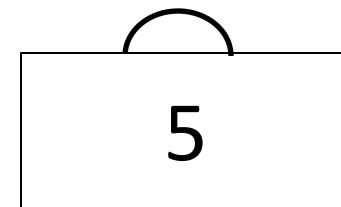


num



If Else Revisited

```
num = int(input("Enter a number: "))  
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    print("Your number is 0 ")  
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    print("Your number is positive")  
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```

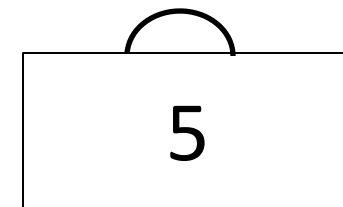


num



If Else Revisited

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    print("Your number is positive")  
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    print("Your number is negative")
```

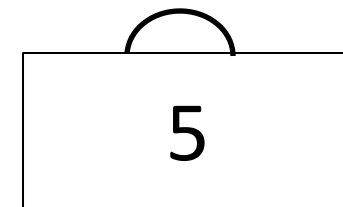
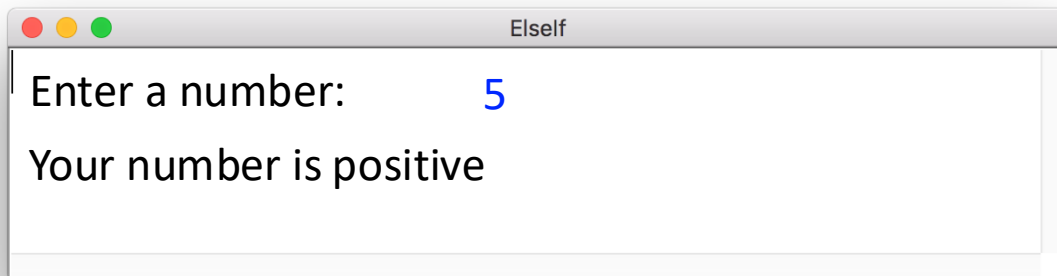


num



If Else Revisited

```
num = int(input("Enter a number: "))  
if num == 0:  
    print("Your number is 0 ")  
elif num > 0:  
    print("Your number is positive")  
else:  
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```

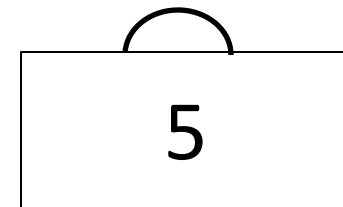
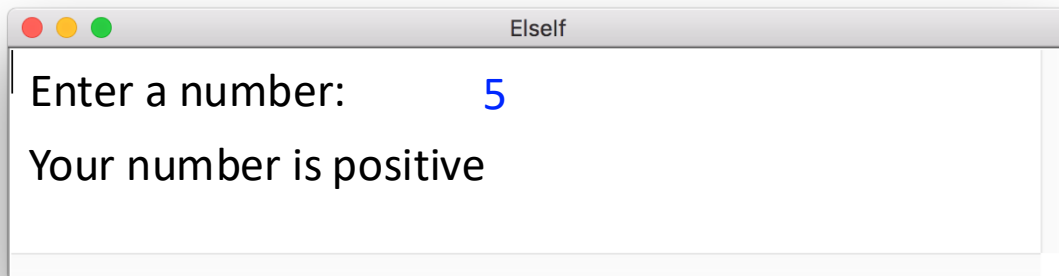
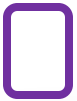


num

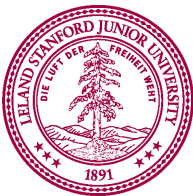


If Else Revisited

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    print("Your number is positive")  
else:  
    print("Your number is negative")
```



num

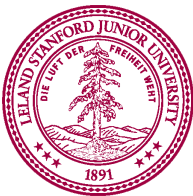


Conditions in Python



Use **while** and **if** statements in Python.

They are the same as in Karel, except that the *test* can be any expression that evaluates to **True** or **False**



Amazing

Guess My Number

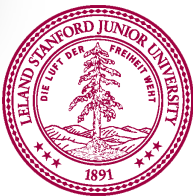
```
GuessMyNumber
I am thinking of a number between 0 and 99...
Enter a guess: 50
Your guess is too high

Enter a new number: 25
Your guess is too low

Enter a new number: 40
Your guess is too low

Enter a new number: 45
Your guess is too low

Enter a new number: 48
Congrats! The number was: 48
|
```

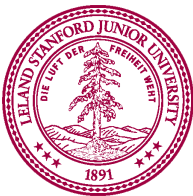
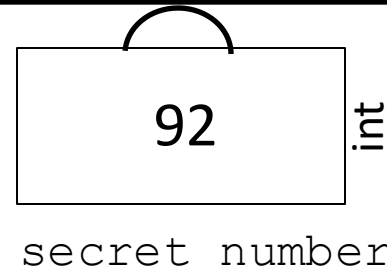


Guess My Number

```
secret_number = random.randint(1, 99)
print("I am thinking of a number between 1 and 99...")
guess = int(input("Enter a guess: "))
# True if guess is not equal to secret number
while guess != secret_number:
    # True if guess is less than secret number
    if guess < secret_number:
        print("Your guess is too low")
    else:
        print("Your guess is too high")

    print("") # an empty line
    guess = int(input("Enter a new guess: "))

print(f"Congrats! The number was: {secret_number}")
```

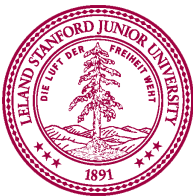
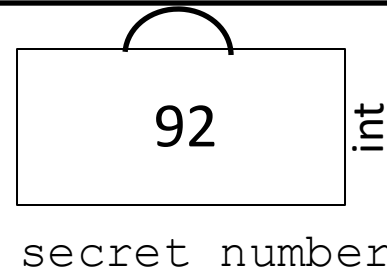


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```

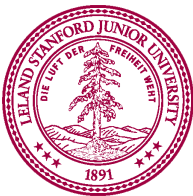
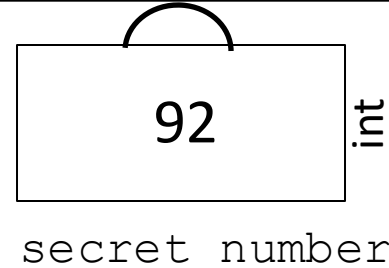
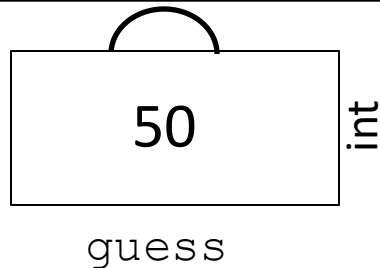


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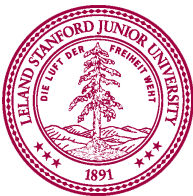
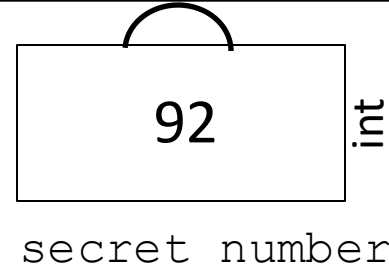
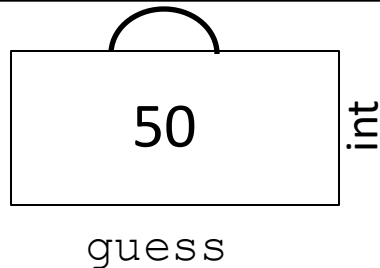


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```

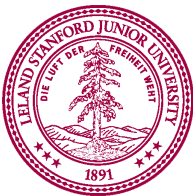
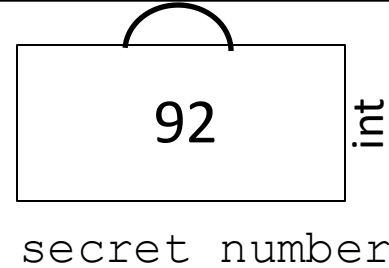
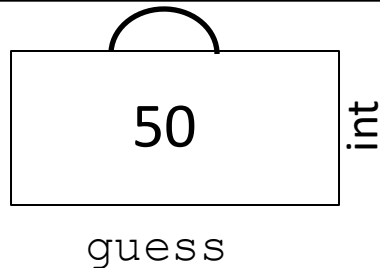


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```

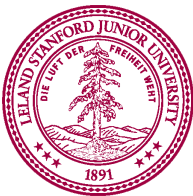
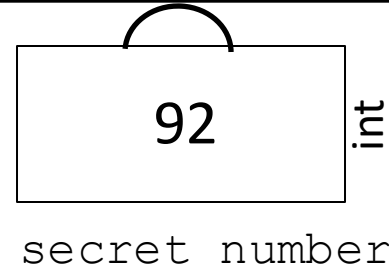
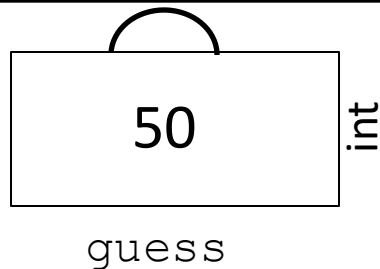


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print(f"Congrats! The number was: {secret_number}")
```

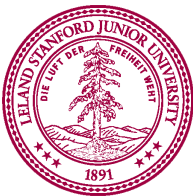
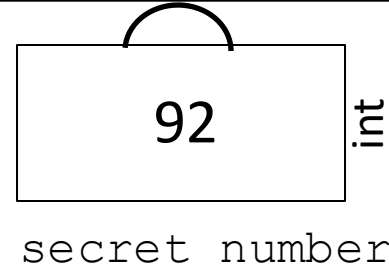
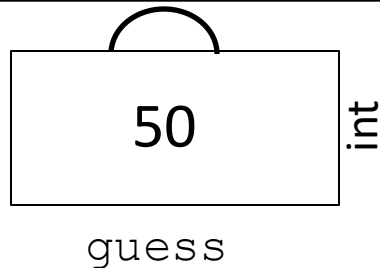


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print(f"Congrats! The number was: {secret_number}")
```

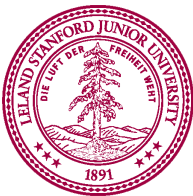
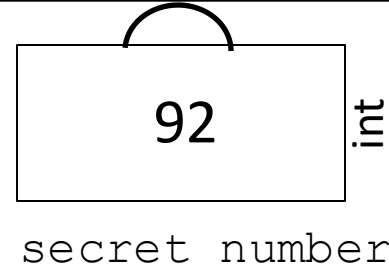
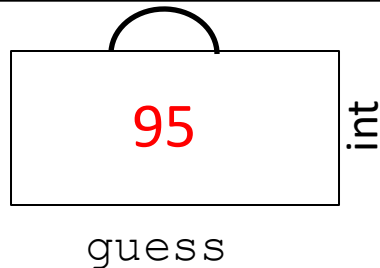


Guess My Number

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secret_number = random.randint(1, 99)
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print(f"Congrats! The number was: {secret_number}")
```

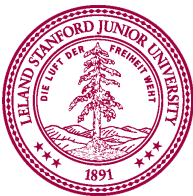
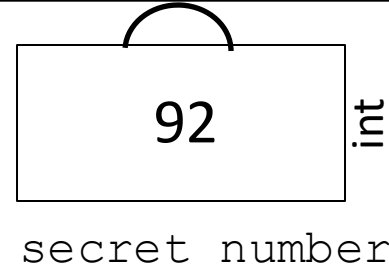
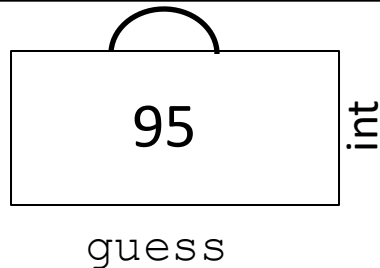


Guess My Number

```
secret_number = random.randint(1, 99)
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guess = int(input("Enter a guess: "))
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    guess = int(input("Enter a new guess: "))


print(f"Congrats! The number was: {secret_number}")
```

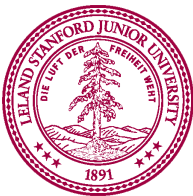
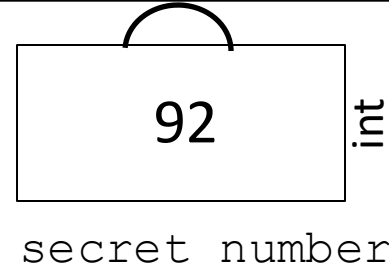
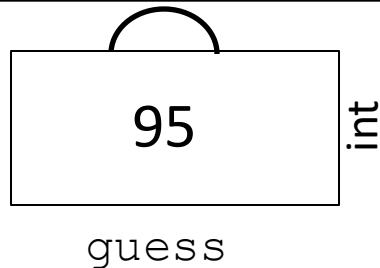


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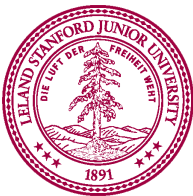
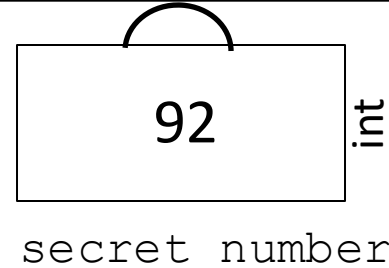
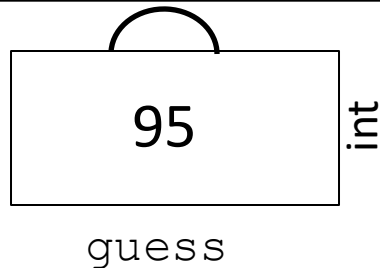


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print(f"Congrats! The number was: {secret_number}")
```

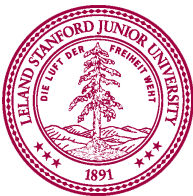
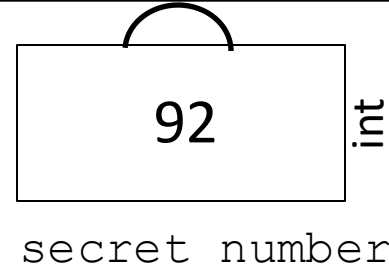
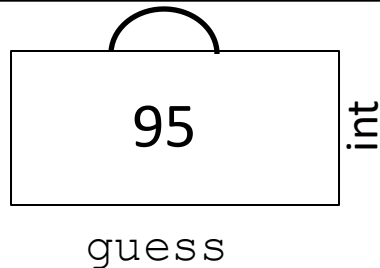


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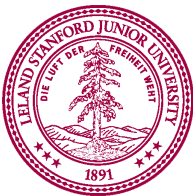
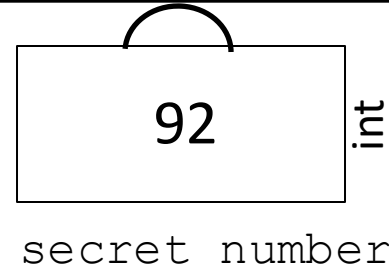
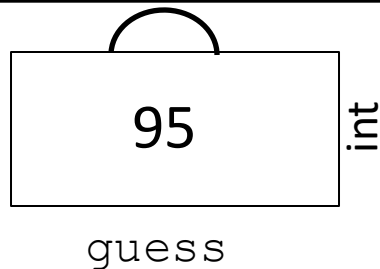


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    else:
        print("Your guess is too high")

    print("") # an empty line
    guess = int(input("Enter a new guess: "))

print(f"Congrats! The number was: {secret_number}")
```

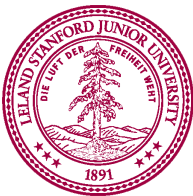
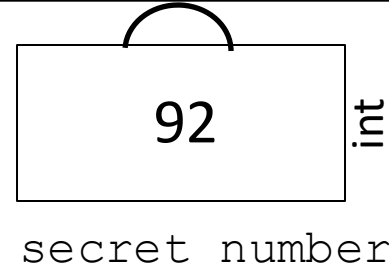
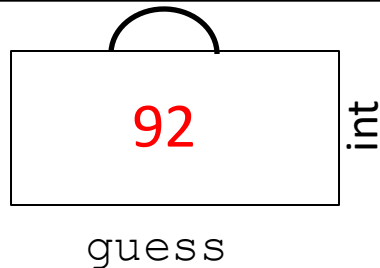


Guess My Number

```
secret_number = random.randint(1, 99)
print("I am thinking of a number between 1 and 99...")
guess = int(input("Enter a guess: "))
# True if guess is not equal to secret number
while guess != secret_number:
    # True if guess is less than secret number
    if guess < secret_number:
        print("Your guess is too low")
    else:
        print("Your guess is too high")

    print("") # an empty line
    guess = int(input("Enter a new guess: "))

print(f"Congrats! The number was: {secret_number}")
```

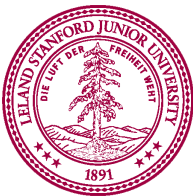
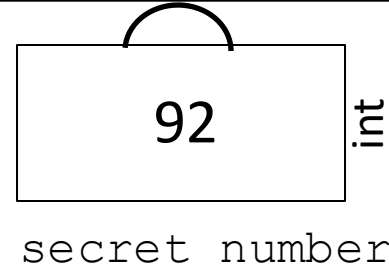
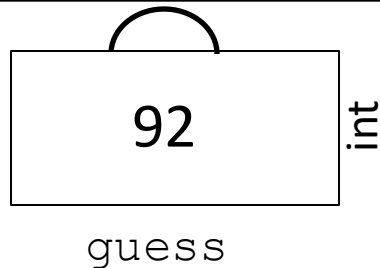


Guess My Number

```
secret_number = random.randint(1, 99)
print("I am thinking of a number between 1 and 99...")
guess = int(input("Enter a guess: "))
# True if guess is not equal to secret number
while guess != secret_number:
    # True if guess is less than secret number
    if guess < secret_number:
        print("Your guess is too low")
    else:
        print("Your guess is too high")

    print("") # an empty line
    guess = int(input("Enter a new guess: "))


print(f"Congrats! The number was: {secret_number}")
```

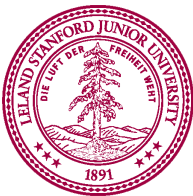
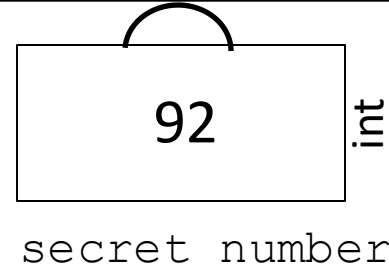
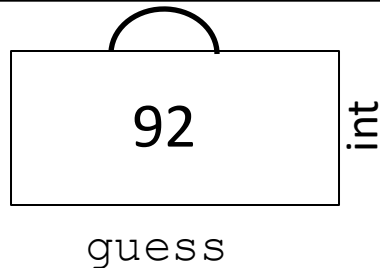


Guess My Number

```
secret_number = random.randint(1, 99)
print("I am thinking of a number between 1 and 99...")
guess = int(input("Enter a guess: "))
# True if guess is not equal to secret number
while guess != secret_number:
    # True if guess is less than secret number
    if guess < secret_number:
        print("Your guess is too low")
    else:
        print("Your guess is too high")

    print("") # an empty line
    guess = int(input("Enter a new guess: "))

print(f"Congrats! The number was: {secret_number}")
```

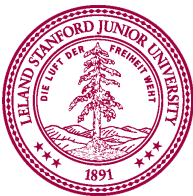
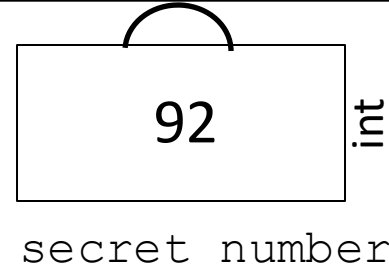
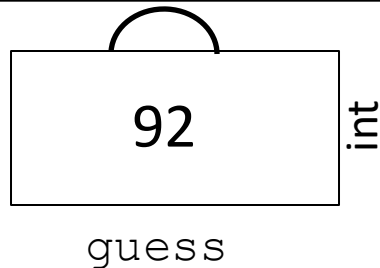


Guess My Number

```
secret_number = random.randint(1, 99)
print("I am thinking of a number between 1 and 99...")
guess = int(input("Enter a guess: "))
# True if guess is not equal to secret number
while guess != secret_number:
    # True if guess is less than secret number
    if guess < secret_number:
        print("Your guess is too low")
    else:
        print("Your guess is too high")

    print("") # an empty line
    guess = int(input("Enter a new guess: "))

print(f"Congrats! The number was: {secret_number}")
```



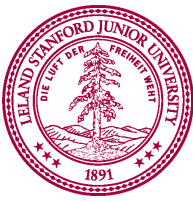
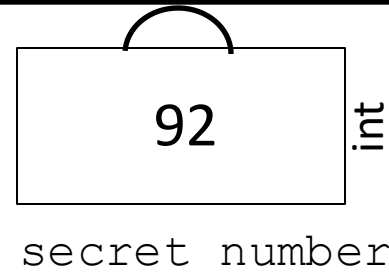
Behind the Scenes



Guess My Number

```
secret_number = random.randint(1, 99)
print("I am thinking of a number between 1 and 99...")
```

```
print(f"Congrats! The number was: {secret_number}")
```

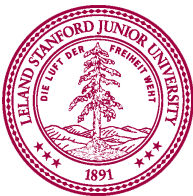


Guess My Number

```
secret_number = random.randint(1, 99)
print("I am thinking of a number between 1 and 99...")
```

```
while ???:  
    # Repeat some stuff???
```

```
print(f"Congrats! The number was: {secret_number}")
```

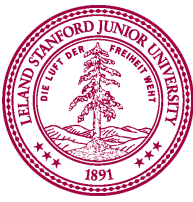


Guess My Number

```
secret_number = random.randint(1, 99)
print("I am thinking of a number between 1 and 99...")
```

```
while ???:  
    # Get a new guess  
  
    # Report high/low
```

```
print(f"Congrats! The number was: {secret_number}")
```



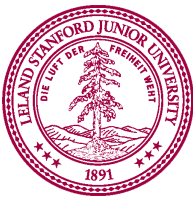
Guess My Number

```
secret_number = random.randint(1, 99)
print("I am thinking of a number between 1 and 99...")
```

```
while ???:  
    # Get a new guess  
    guess = int(input("Enter a guess: "))
```

```
    # Report high/low
```

```
print(f"Congrats! The number was: {secret_number}")
```



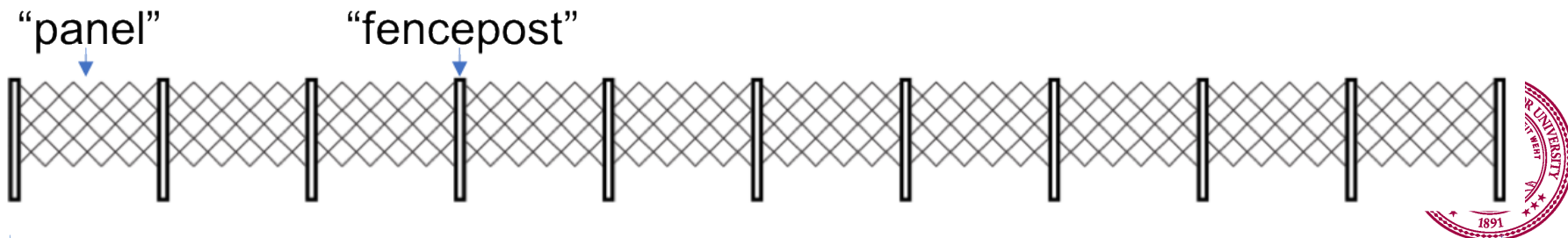
Guess My Number

```
secret_number = random.randint(1, 99)
print("I am thinking of a number between 1 and 99...")
```

```
while guess != secret_number:
    # Get a new guess
    guess = int(input("Enter a guess: "))
```

```
    # Report high/low
```

```
print(f"Congrats! The number was: {secret_number}")
```



Guess My Number

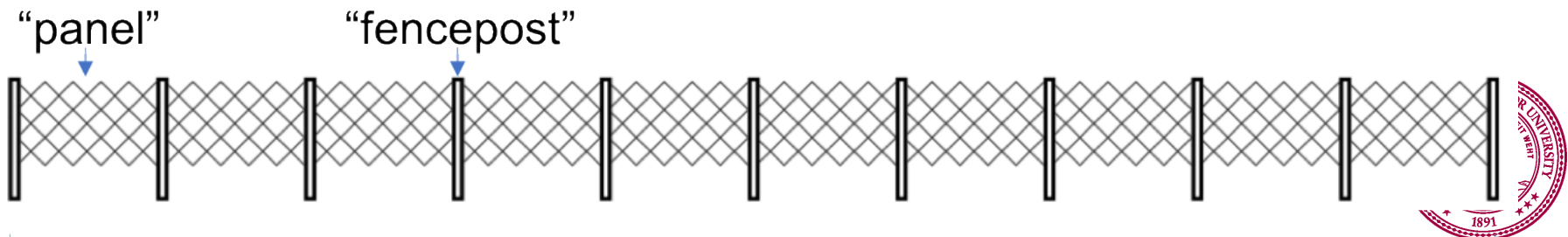
```
secret_number = random.randint(1, 99)
print("I am thinking of a number between 1 and 99...")
guess = int(input("Enter a guess: "))
```

```
while guess != secret_number:
    # Report high/low
```

```
# Get a new guess
```

```
    guess = int(input("Enter a guess: "))
```

```
print(f"Congrats! The number was: {secret_number}")
```



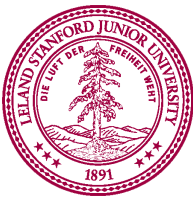
Guess My Number

```
secret_number = random.randint(1, 99)
print("I am thinking of a number between 1 and 99...")
guess = int(input("Enter a guess: "))

while guess != secret_number:
    # True if guess is less than secret number
    if guess < secret_number:
        print("Your guess is too low")
    else:
        print("Your guess is too high")

    print("") # an empty line
    guess = int(input("Enter a new guess: "))

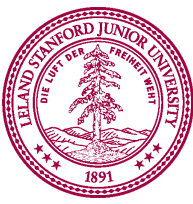
print(f"Congrats! The number was: {secret_number}")
```



George Boole



English Mathematician teaching in Ireland 1815 – 1864
Boole died of being too cool



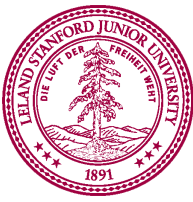
Logical Operators

In order of precedence:

Operator	Example	Result
<code>not</code>	<code>not (2 == 3)</code>	<code>True</code>
<code>and</code>	<code>(2 == 3) and (-1 < 5)</code>	<code>False</code>
<code>or</code>	<code>(2 == 3) or (-1 < 5)</code>	<code>True</code>

Can "chain" tests

```
# is x between 2 and 10?  
2 <= x and x <= 10
```



Boolean Variables

```
# Store expressions that evaluate to True/False
```

```
x = 1 < 2      # True
```

```
y = 5.0 == 4.0 # False
```

```
# Directly set to True/False
```

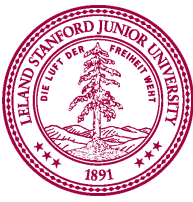
```
is_sheltering = True
```

```
is_raining = False
```

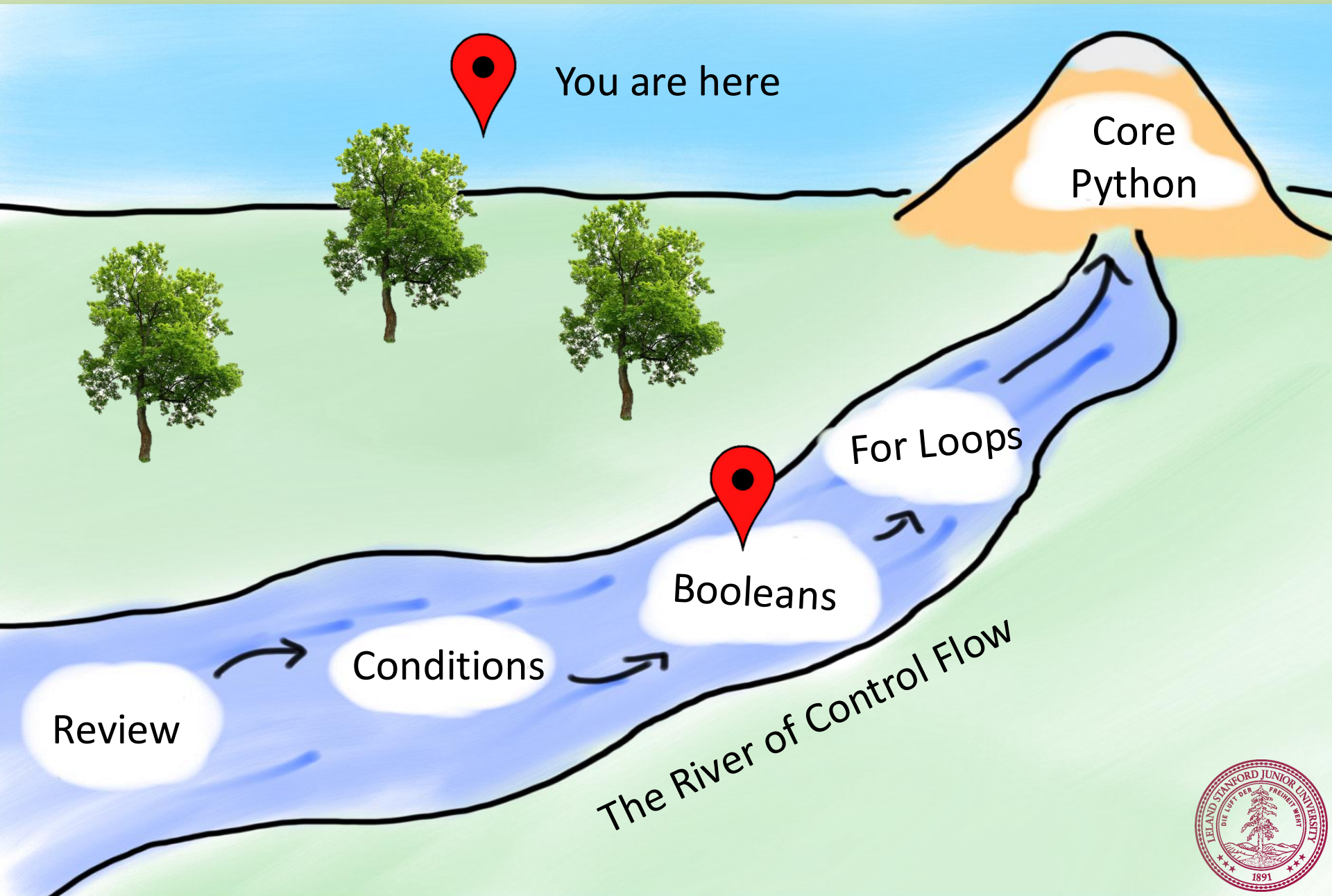
```
play_again = input('Play again? "y" or "n"') == 'y'
```

```
if play_again:
```

```
    ...
```



Today's Route

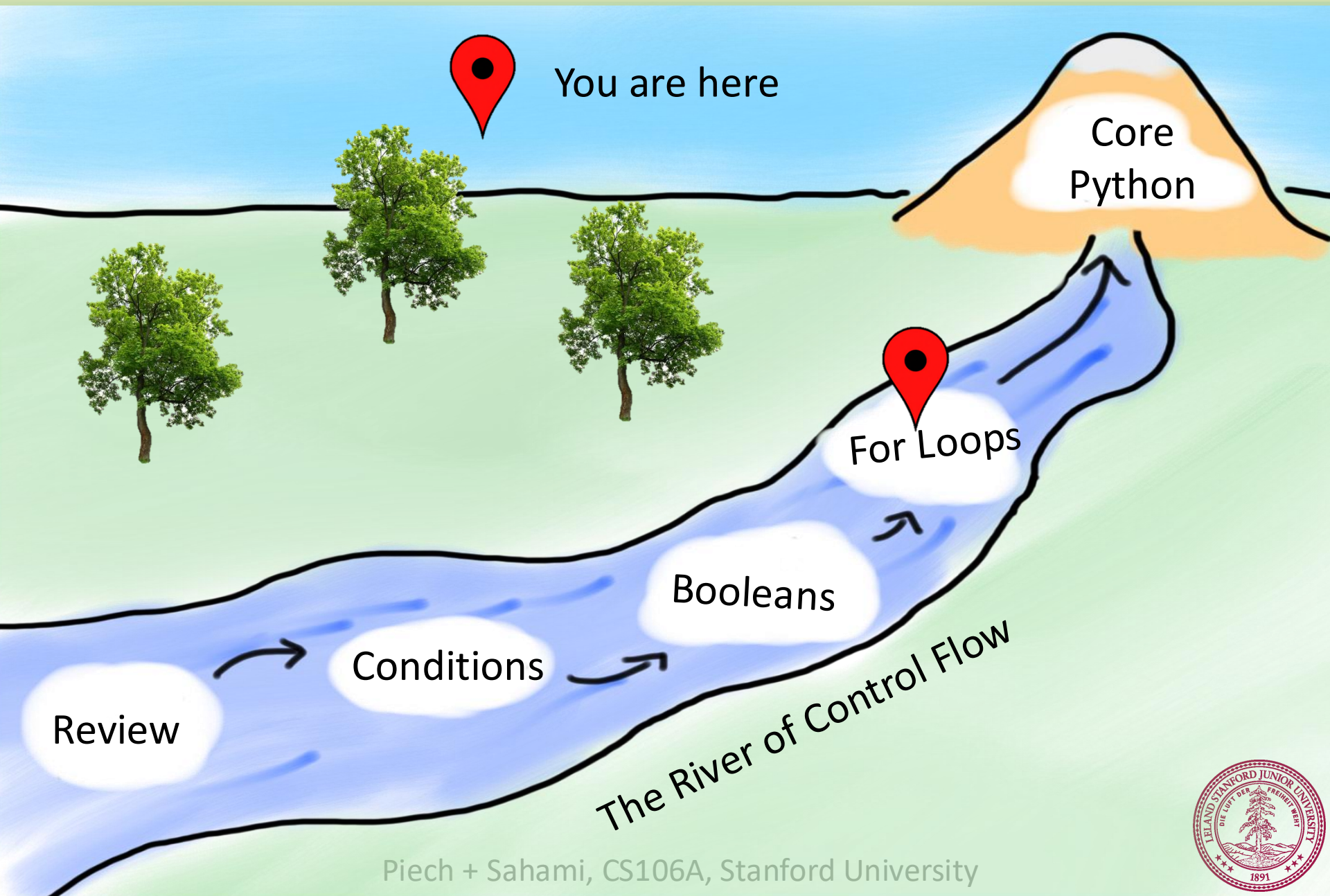




Control Flow Revisited

Chris Piech and Mehran Sahami
Stanford University

Today's Route



How would you print “Python rocks socks”
100 times

For Loop Redux

```
def main():  
    for i in range(100):  
        print("Python rocks socks!")
```



For Loop Redux

```
for i in range(100):  
    print("Python rocks socks!")
```

```
i = 0  
while i < 100:  
    print("Python rocks socks!")  
    i = i + 1
```

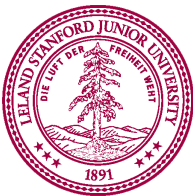


For Loop Redux

Create a counting
variable i

Which takes on the
values 0 to 99 one at
a time

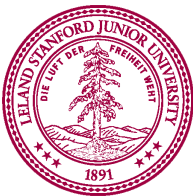
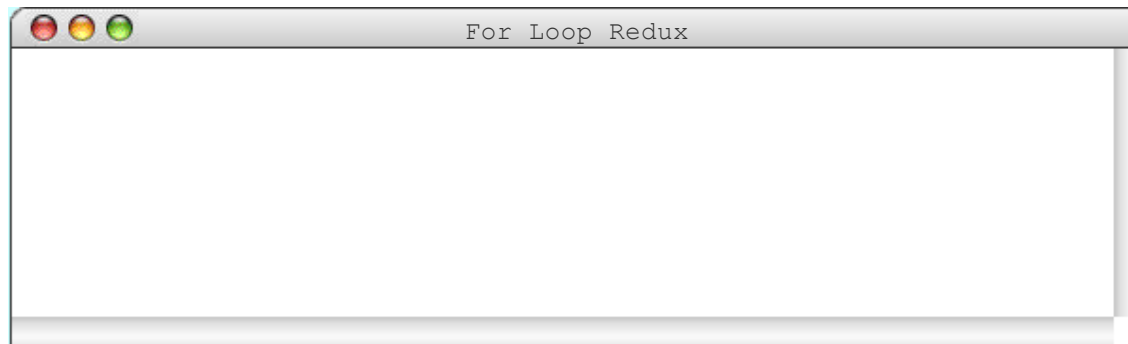
```
for i in range(100):  
    print("Python rocks socks!")
```



For Loop Redux

range(3) -> [0, 1, 2]

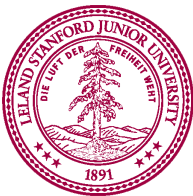
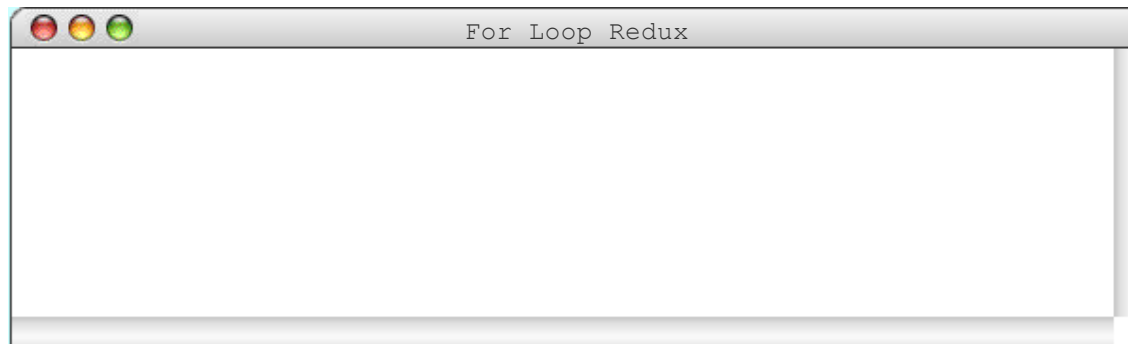
```
for i in range(3):  
    print("Python rocks socks!")
```



For Loop Redux

range(3) -> [0, 1, 2]

```
for i in range(3):  
    print("Python rocks socks!")
```

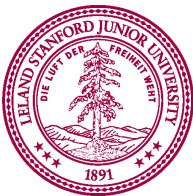
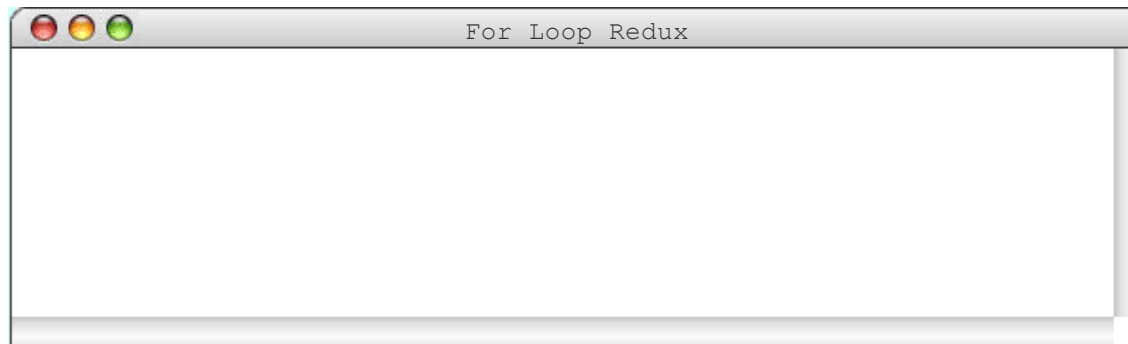


For Loop Redux

i 0

range(3) -> [0, 1, 2]

```
for i in range(3):  
    print("Python rocks socks!")
```



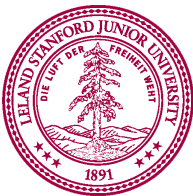
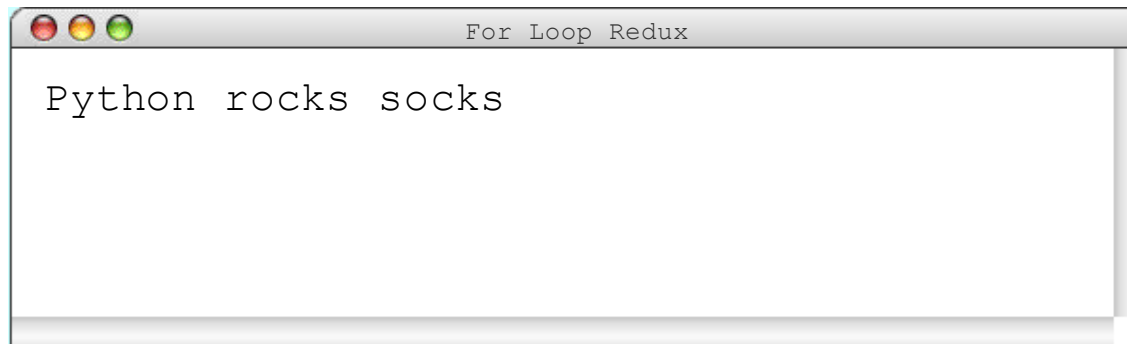
For Loop Redux

i 0

range(3) -> [0, 1, 2]

```
for i in range(3):
```

```
    print("Python rocks socks!")
```

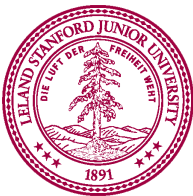
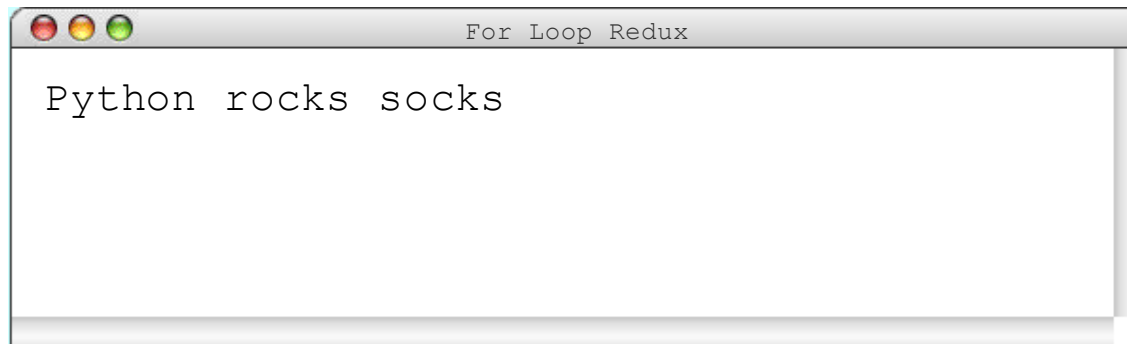


For Loop Redux

i 0

range(3) -> [0, 1, 2]

```
for i in range(3):  
    print("Python rocks socks!")
```

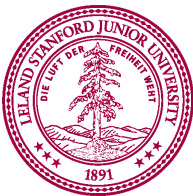
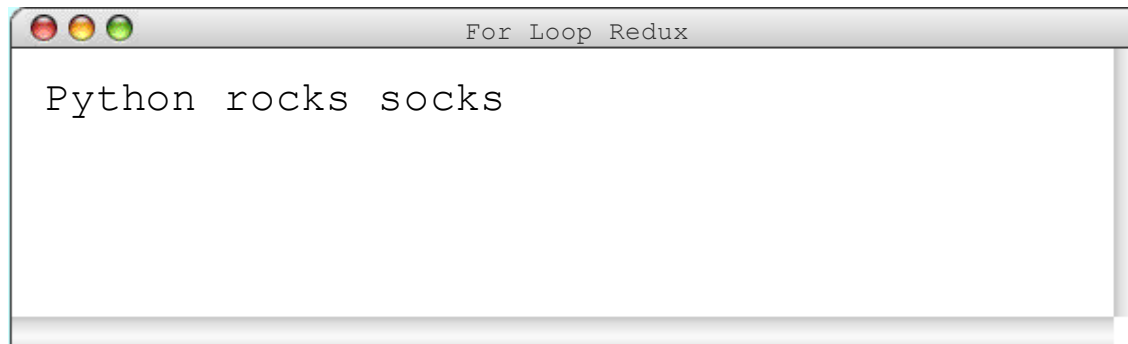


For Loop Redux

i 1

range(3) -> [0, 1, 2]

```
for i in range(3):  
    print("Python rocks socks!")
```



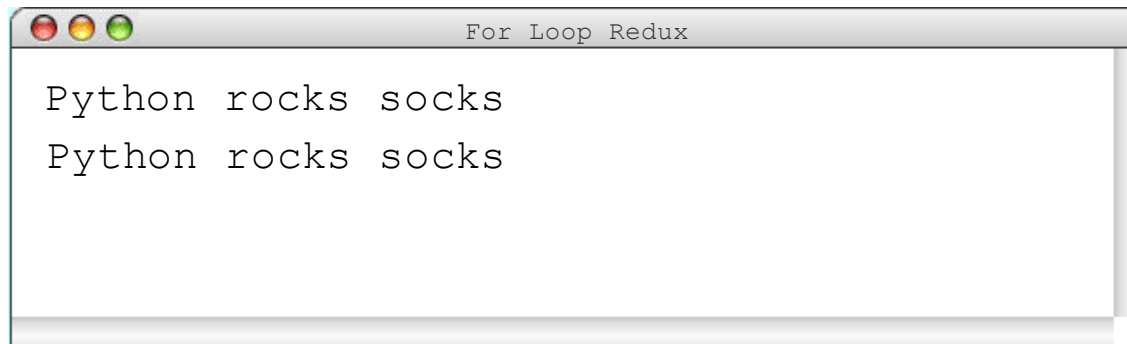
For Loop Redux

i 1

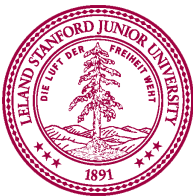
range(3) -> [0, 1, 2]

```
for i in range(3):
```

```
    print("Python rocks socks!")
```



```
For Loop Redux  
Python rocks socks  
Python rocks socks
```

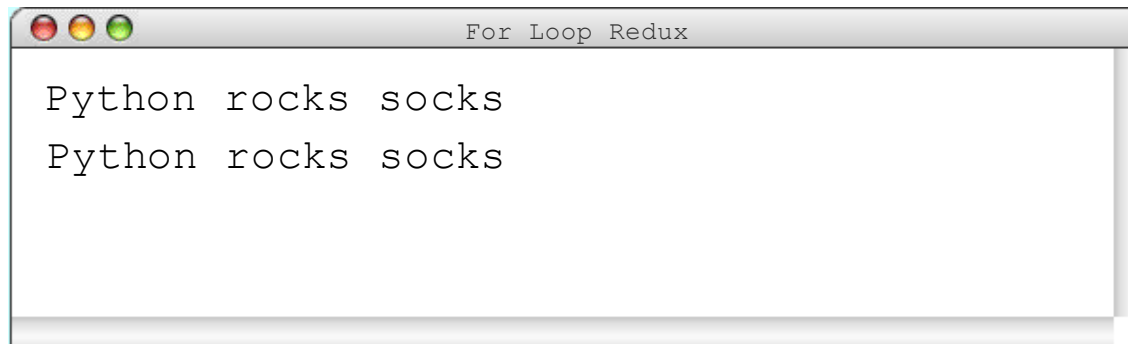


For Loop Redux

i 1

range(3) -> [0, 1, 2]

```
for i in range(3):  
    print("Python rocks socks!")
```

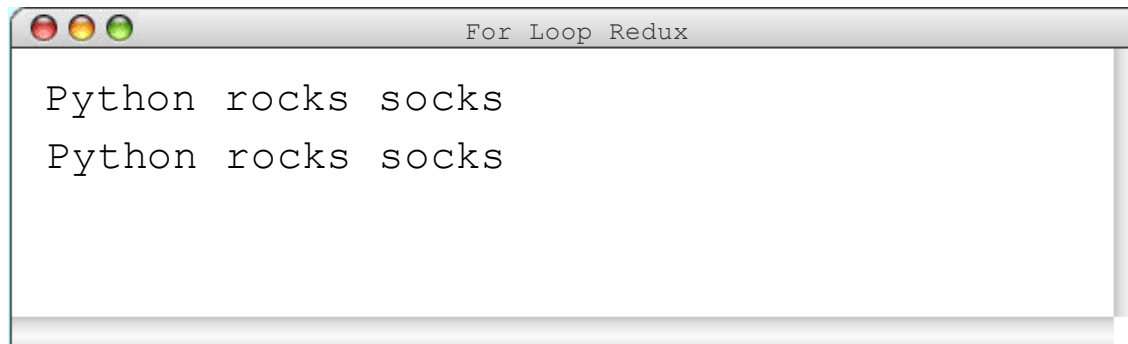


For Loop Redux

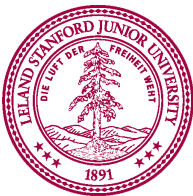
i 2

range(3) -> [0, 1, 2]

```
for i in range(3):  
    print("Python rocks socks!")
```



```
For Loop Redux  
  
Python rocks socks  
Python rocks socks
```



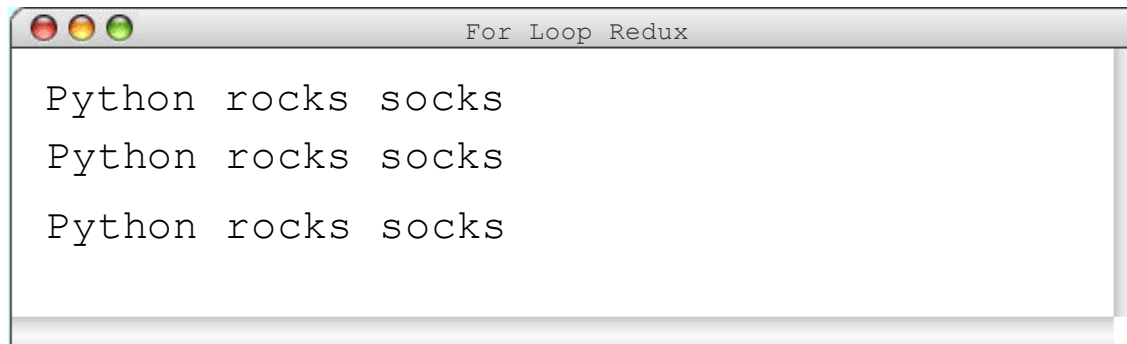
For Loop Redux

i 2

range(3) -> [0, 1, 2]

for i **in** range(3):

print("Python rocks socks!")



A terminal window titled "For Loop Redux" with three colored window control buttons (red, yellow, green) in the top-left corner. The terminal displays the output of the code above, which is the string "Python rocks socks" repeated on three separate lines.

```
Python rocks socks
Python rocks socks
Python rocks socks
```



For Loop Redux

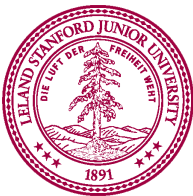
i 2

range(3) -> [0, 1, 2]

```
for i in range(3):  
    print("Python rocks socks!")
```



```
For Loop Redux  
  
Python rocks socks  
Python rocks socks  
Python rocks socks
```

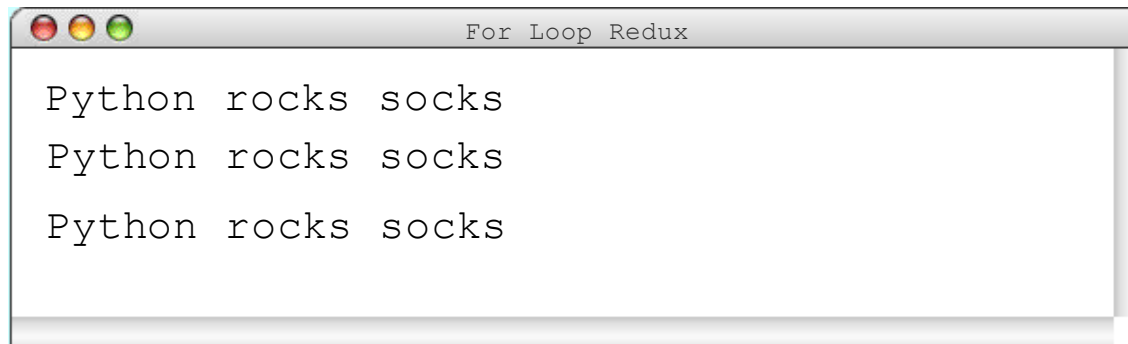


For Loop Redux

i 3

range(3) -> [0, 1, 2]

```
for i in range(3):  
    print("Python rocks socks!")
```



```
For Loop Redux  
  
Python rocks socks  
Python rocks socks  
Python rocks socks
```

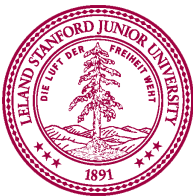
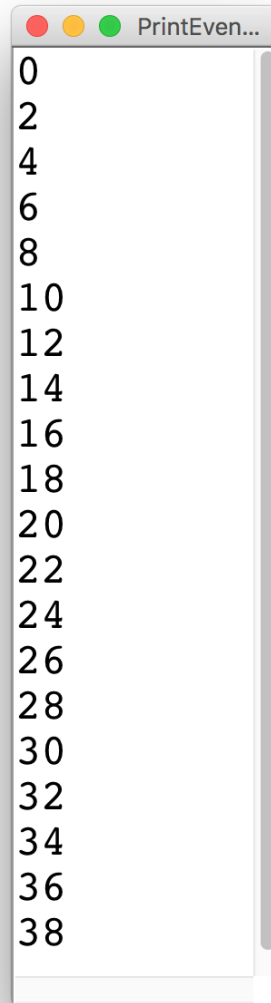


You can use the for loop variable



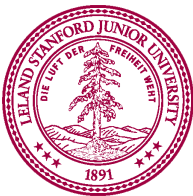
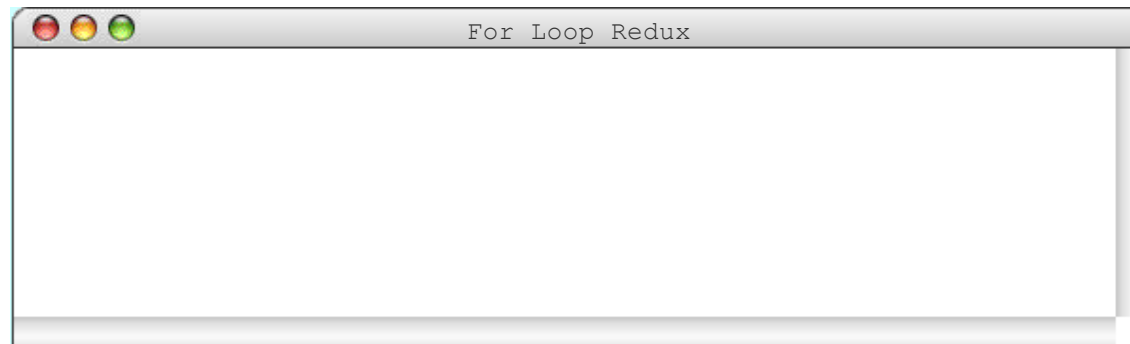
How would you print the first 100 even numbers?

Printing Even Numbers



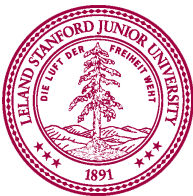
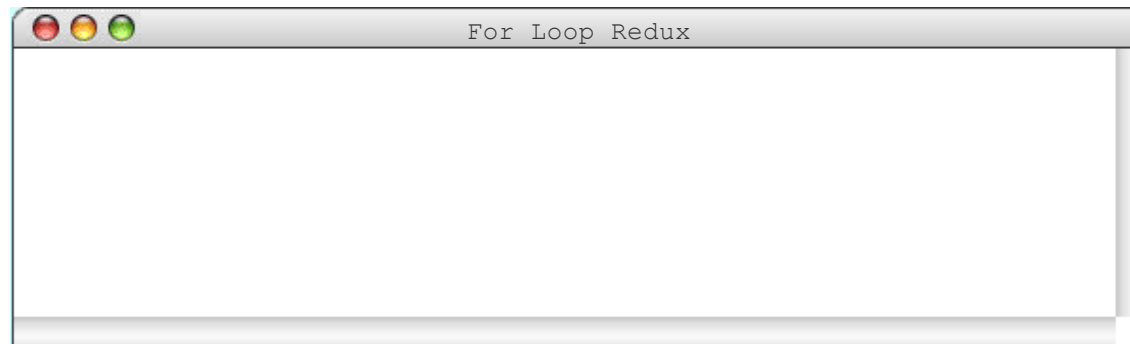
Printing Even Numbers

```
for i in range(3):  
    print(i * 2)
```



Printing Even Numbers

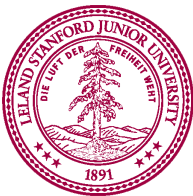
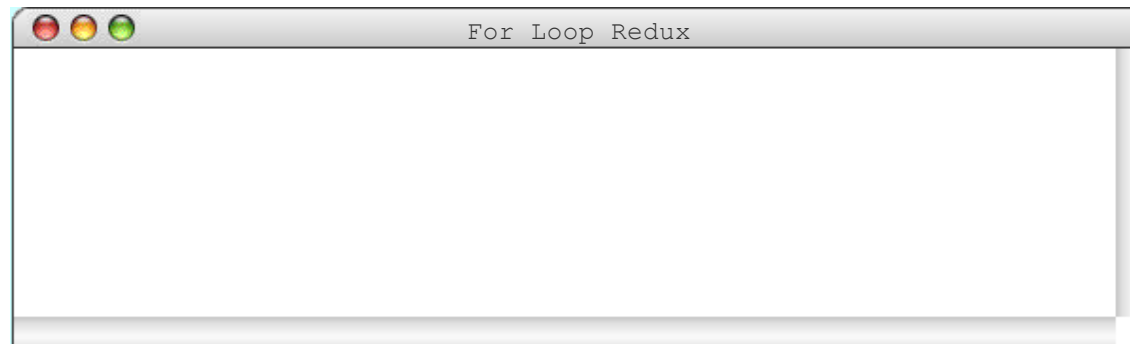
```
for i in range(3):  
    print(i * 2)
```



Printing Even Numbers

i 0

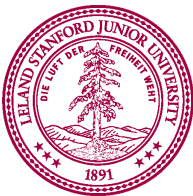
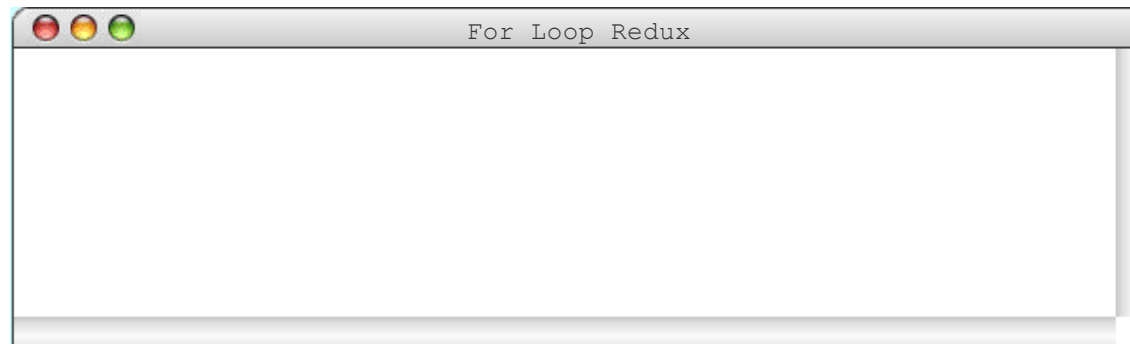
```
for i in range(3):  
    print(i * 2)
```



Printing Even Numbers

i 0

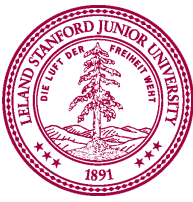
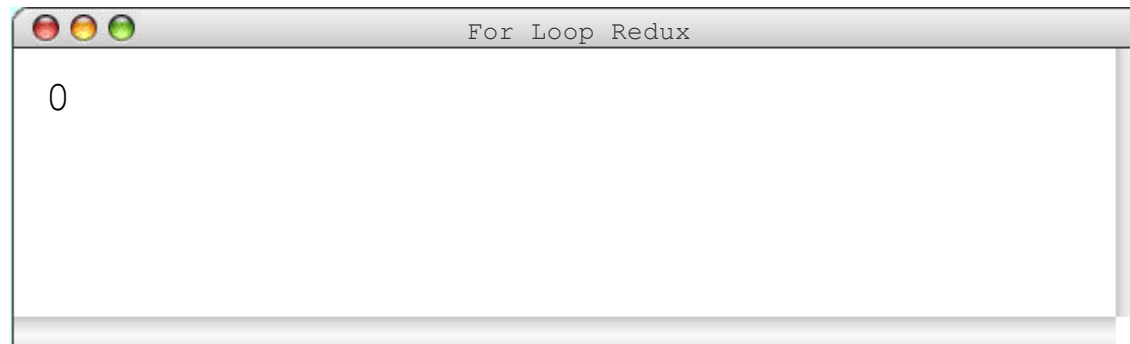
```
for i in range(3):  
    print(i * 2)
```



Printing Even Numbers

i 0

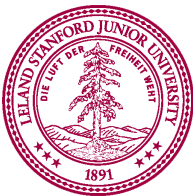
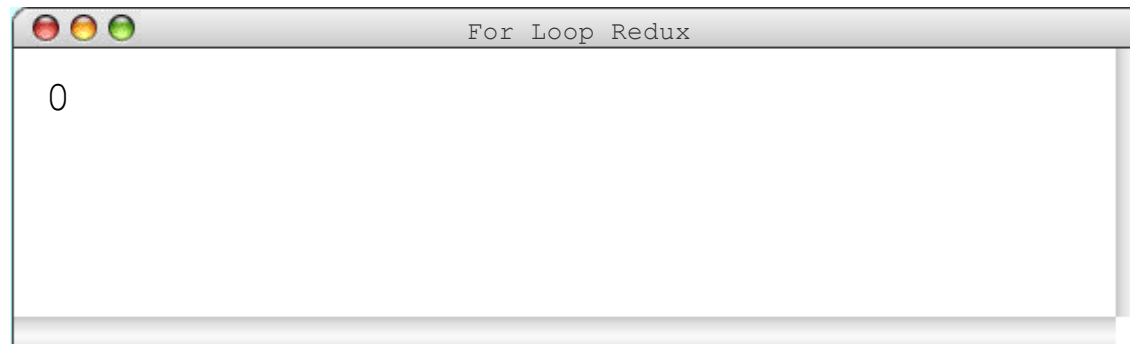
```
for i in range(3):  
    print(i * 2)
```



Printing Even Numbers

i 1

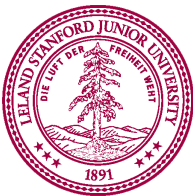
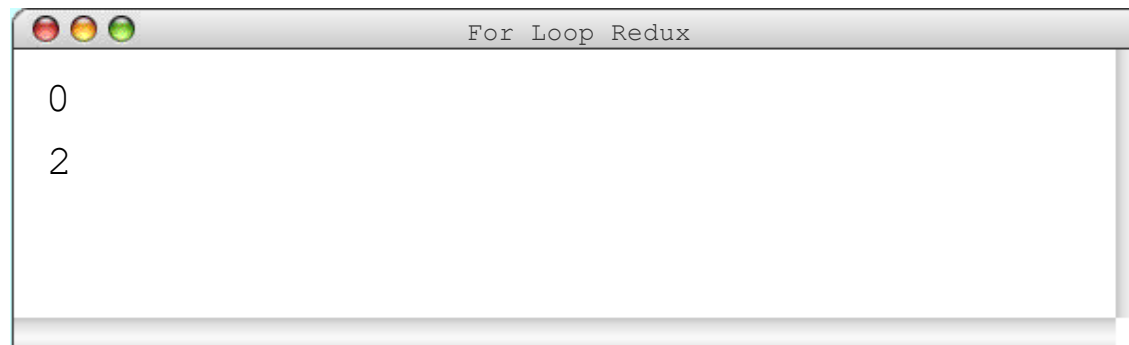
```
for i in range(3):  
    print(i * 2)
```



Printing Even Numbers

i 1

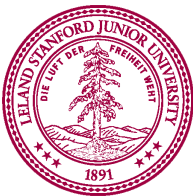
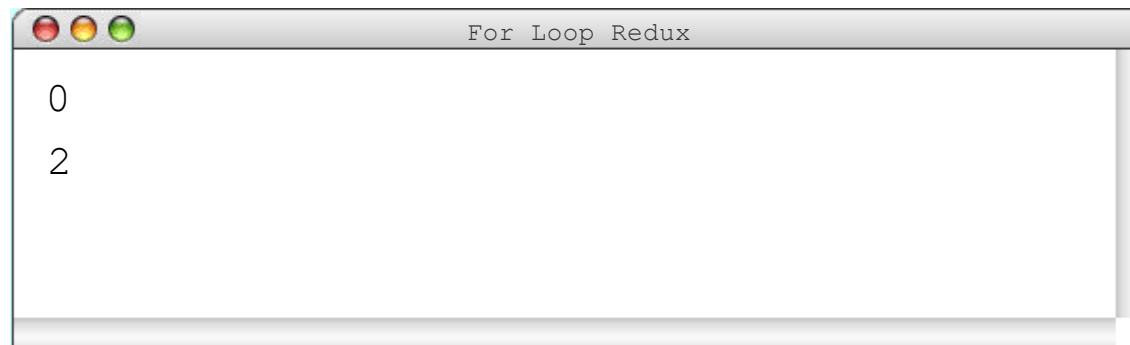
```
for i in range(3):  
    print(i * 2)
```



Printing Even Numbers

i 2

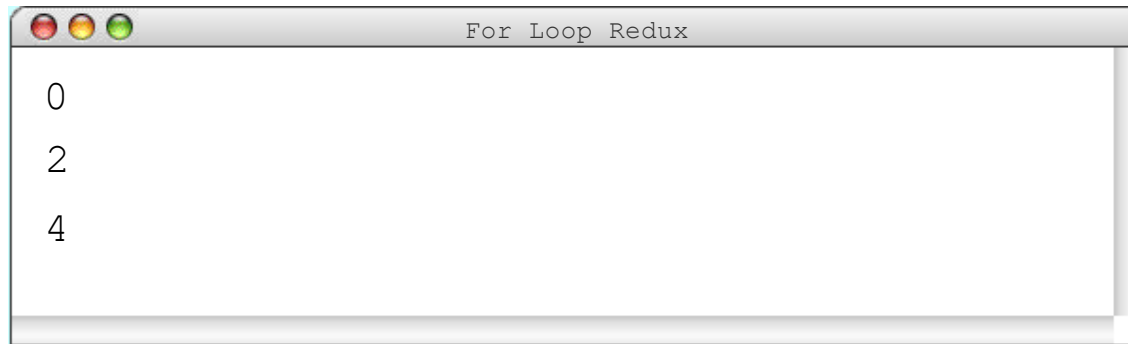
```
for i in range(3):  
    print(i * 2)
```



Printing Even Numbers

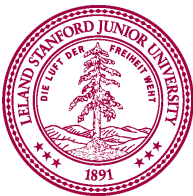
i 2

```
for i in range(3):  
    print(i * 2)
```



A terminal window titled "For Loop Redux" showing the output of the Python code. The output consists of three lines: 0, 2, and 4, each on a new line.

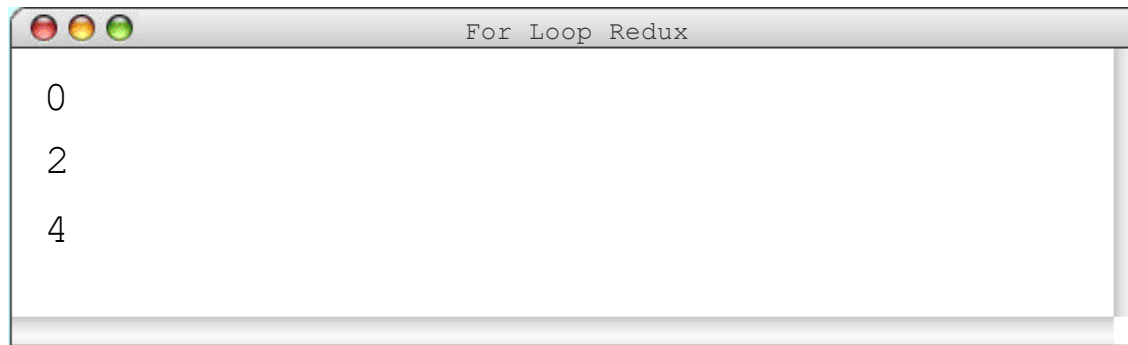
```
0  
2  
4
```



Printing Even Numbers

i 3

```
for i in range(3):  
    print(i * 2)
```



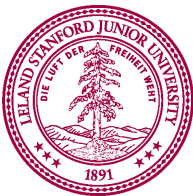
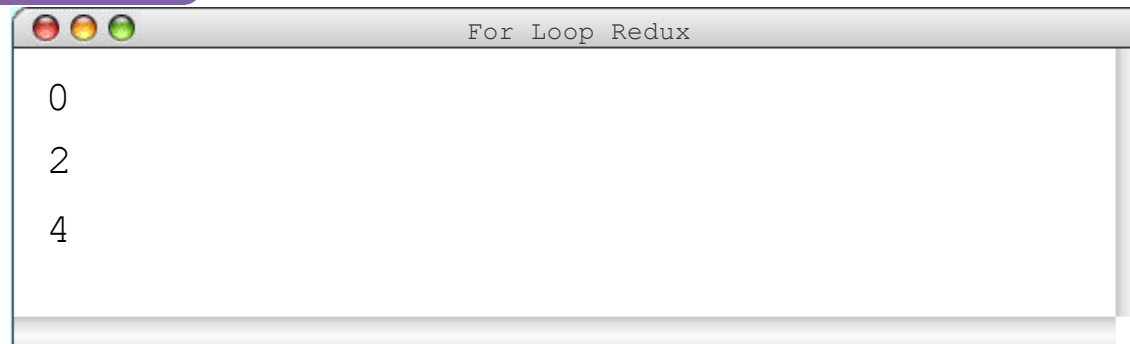
```
For Loop Redux  
0  
2  
4
```



Printing Even Numbers

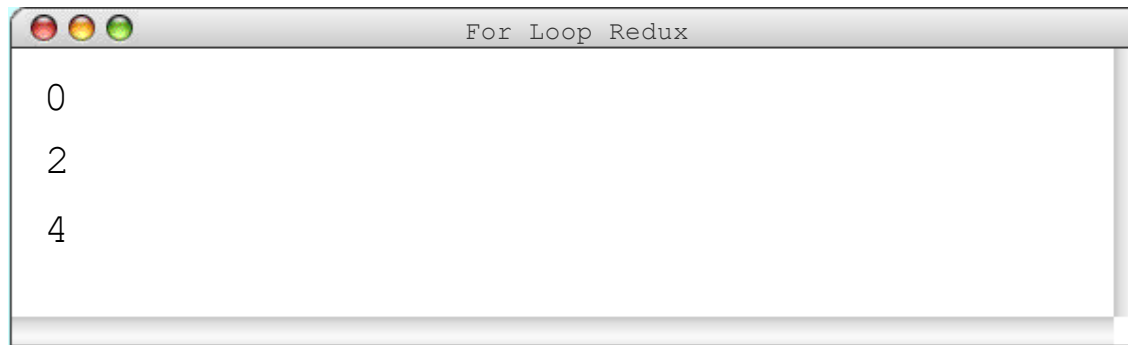
i 3

```
for i in range(3):  
    print(i * 2)
```



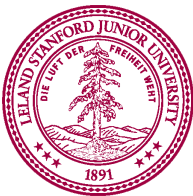
Printing Even Numbers

```
for i in range(3):  
    print(i * 2)
```



A terminal window titled "For Loop Redux" showing the output of the Python code. The output consists of three lines: 0, 2, and 4, each on a new line.

```
0  
2  
4
```



Printing Even Numbers

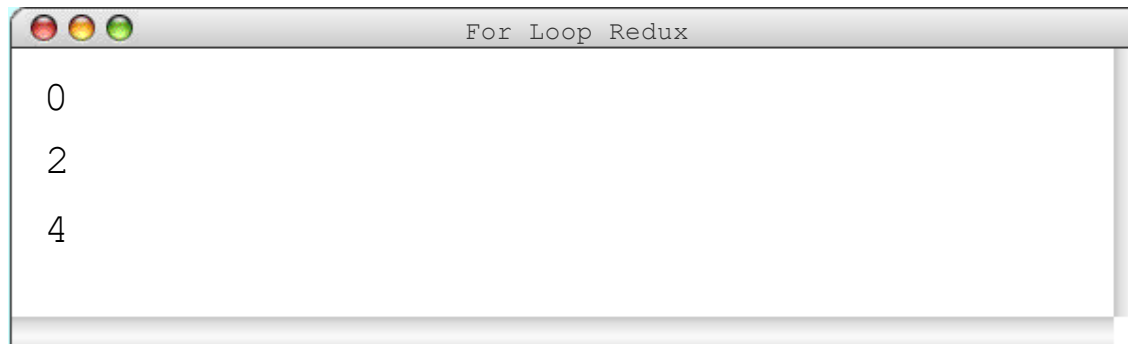
```
# our solution    0,1,2
for i in range(3):
    print(i * 2)
```

```
# equivalently
for i in range(0, 6, 2):
    print(i)
```

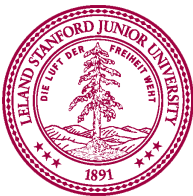
Start at 0

Stop before 6

Skip by 2 each time

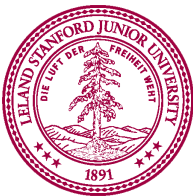


```
For Loop Redux
0
2
4
```

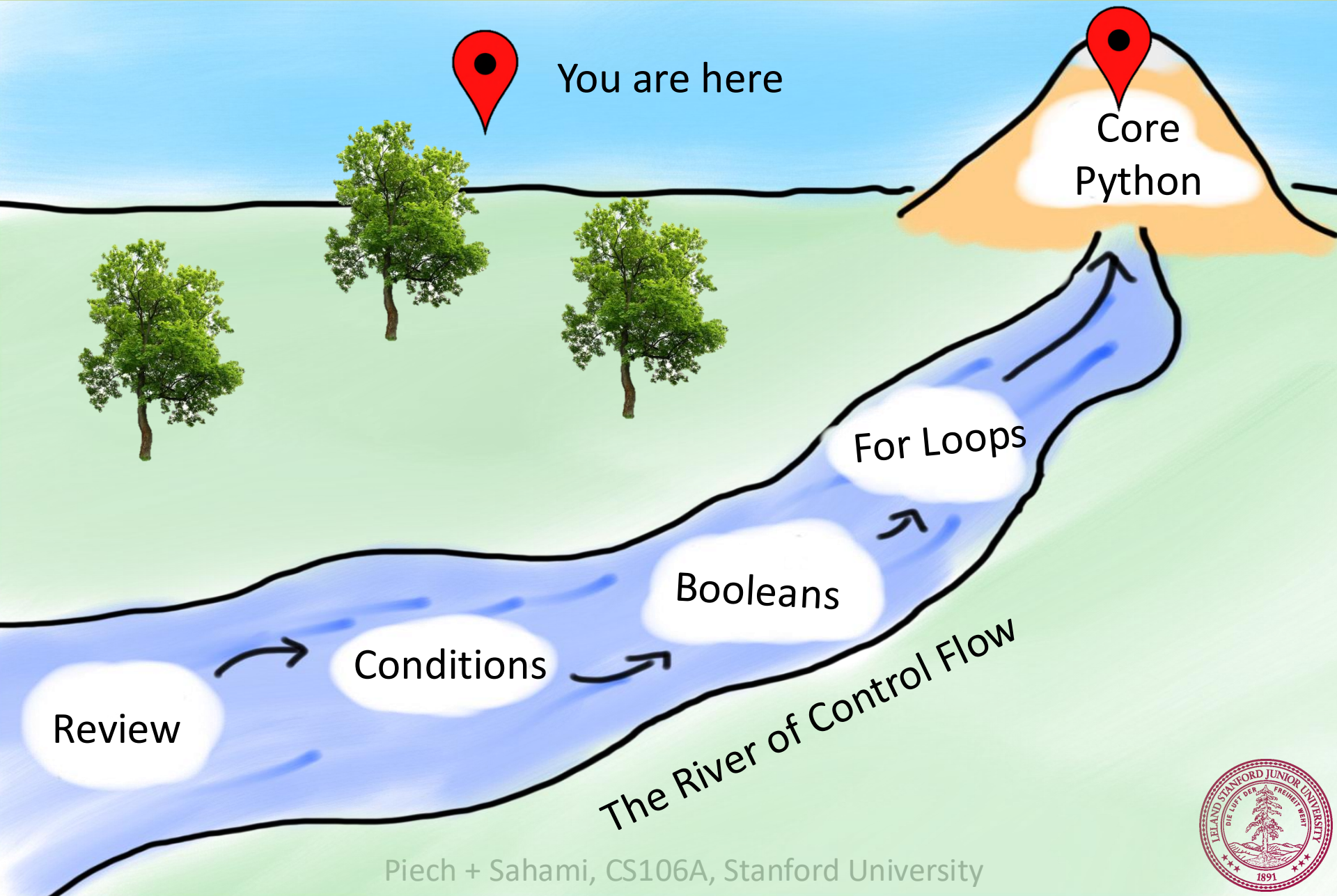


That's all

```
def main() :  
    for i in range(999999) :  
        print("You rock!")  
        print("Keep learning!")
```



Today's Route



Today's Goal

1. Be able to use For / While / If in Python

